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Comparing Kappa Weighting and Causal Machine Learning Estimators via Weight-Based Diagnostics

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Abstract

This thesis studies classical kappa estimators of the LATE together with machine-learning-assisted IV estimators, focusing on Wald-AIPW and PLR-IV. It combines **?** kappa framework with recent results on estimator normalization and outcome weights to place both estimator classes within a common diagnostic framework. In particular, the thesis constructs and implements outcome weights for five kappa-based estimators and uses them to examine normalization, covariate balance, weight concentration, and sensitivity to outcome transformations.

A central contribution is the extension of translation-invariance analysis from classical kappa estimators to complete data-adaptive estimation procedures. Rather than evaluating only a fixed set of fitted weights, the analysis reruns nuisance estimation, cross-fitting, and hyperparameter tuning after economically irrelevant changes in the coding or units of the outcome. The empirical applications revisit the Vietnam draft lottery, Card's college-proximity design, and the Angrist–Evans fertility design, and compare normalized and unnormalized kappa estimators with PLR-IV and Wald-AIPW implemented using several nuisance learners.

The empirical analysis examines whether normalization is reflected in translation invariance and whether outcome-weight diagnostics reveal differences in effective support, concentration, negative weights, and covariate balance that are not visible from point estimates alone.

Keywords: local average treatment effect, instrumental variables, kappa weighting, outcome weights, translation invariance, covariate balance, double machine learning.

JEL classification: C14, C21, C26.

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1 Introduction

Causal estimates should not depend on an arbitrary choice of measurement units. A change from dollars to cents therefore translates every logged outcome by the same constant without changing the underlying economic comparison. An estimator of an effect on log wages or income should consequently be unchanged. Yet this requirement need not hold for all finite-sample weighting estimators of the local average treatment effect (LATE). **?** show that several consistent but unnormalized estimators can react strongly to the centering of the outcome and to the monetary unit chosen before taking logarithms. Normalization is therefore not merely a presentational choice. It can determine whether an estimate respects an economically irrelevant transformation of the data.

This issue is particularly relevant when instrument validity is conditional on observed covariates. Applied work commonly handles such settings with additive two-stage least squares (2SLS). A clean LATE interpretation, however, requires the covariate specification to capture the relevant conditional structure of instrument assignment (**??**). The kappa theorem of **?** offers a flexible alternative. It represents moments of the latent complier population through observed-data weights that depend on the conditional instrument propensity score. The same population identification result gives rise to several sample estimators because alternative expressions for the complier share need not coincide in finite samples. These estimators can target the same population parameter while assigning different coefficients to the observed outcomes.

Flexible estimation of the required conditional relationships has become possible within the double machine learning (DML) framework of **?**. Orthogonal scores and cross-fitting allow nuisance functions to be estimated with parametric models or machine-learning methods. They do not by themselves reveal how a fitted multi-step procedure combines the observations in a realized sample. The outcome-weight framework of **?** supplies this missing finite-sample perspective. It represents a fitted estimate as a weighted sum of the observed outcomes. This common representation makes classical kappa estimators, partially linear IV estimators, and augmented estimators comparable at the level of the sample that produces the reported estimate.

1.1 Related literature

The starting point is the interpretation of instrumental-variable estimands under heterogeneous treatment effects. **?** and **?** establish conditions under which a binary instrument identifies the average treatment effect for compliers. **?** extend this perspective from the complier mean to the marginal distributions of potential outcomes for compliers. Their analysis also demonstrates that conventional IV procedures implicitly construct weighted outcome distributions. **?** then develops a general semiparametric representation for complier moments when instrument assignment is valid conditional on covariates. His kappa weights make the complier population statistically accessible even though individual compliance types are unobserved.

Subsequent work translated conditional LATE identification into flexible estimators. **?** develops regression and weighting procedures that adjust the instrument comparison for covariates. **?** derives a nonparametric estimator based on the ratio of weighted reduced-form and first-stage contrasts. This unnormalized ratio became the main weighting construction in the subsequent methodological literature, although weighting estimators of the LATE remain uncommon in empirical practice (**?**). **?** proposes a separately normalized version of this estimator. More generally, Abadie's alternative representations of the complier share generate several consistent sample analogues. The resulting estimator multiplicity is a finite-sample issue because the corresponding population denominators are equal under the identifying assumptions.

Normalization has a longer history in survey sampling and inverse-probability weighting (**??**). Its role in conditional IV estimation is linked closely to the estimation of the instrument propensity score. Covariate-balancing methods choose propensity-score parameters to satisfy balance conditions rather than only to fit instrument assignment (**?**). **?** adapts this principle to LATE estimation and obtains exact finite-sample balance across instrument groups together with normalized treatment-effect estimates. Building on this literature, **?** provide a unified analysis of five kappa-based LATE estimators. They show that only two are fully normalized and use translation invariance and logarithmic scale invariance as criteria for distinguishing the normalized from the unnormalized estimators. They also show that the normalized estimator proposed by **?** retains useful denominator properties under either form of one-sided noncompliance. Their simulations and three empirical applications motivate its use as the principal normalized benchmark in this thesis.

A parallel literature studies what the implied weights of an estimator reveal about its finite-sample behavior. **?** derive observation-level weights for linear regression and use them to examine covariate balance, representativeness, extrapolation, and effective support. **?** provides a general construction for pseudo-IV estimators and derives outcome weights for prominent DML and generalized-random-forest estimators. His results show that implementation choices determine whether exact outcome weights are available and which normalization properties they satisfy. In particular, partially linear estimators may have zero total weight mass without assigning masses of one and minus one to the treated and untreated components. The framework also permits diagnostic tools developed for explicit weighting estimators to be applied to causal machine-learning procedures.

The term "weight" refers to different objects in these literatures, and these objects must be separated for the empirical analysis below. Abadie's kappa weights are identification objects that recover moments of the complier population. Balancing weights are chosen to satisfy prespecified covariate moments. Outcome weights are the realized observation-level coefficients that reproduce a fitted estimate as a weighted sum of the observed outcomes. They also differ from effect weights, which describe how an estimator aggregates heterogeneous causal effects under additional population assumptions (?). This thesis is concerned primarily with outcome weights. Instrument-balancing weights enter separately when the conditional instrument propensity score is assessed.

1.2 Contribution and empirical approach

This thesis combines the normalization analysis of ? with the outcome-weight framework of ?. It applies this combination to classical and data-adaptive LATE estimators. Its main question is what observation-level outcome weights reveal about the finite-sample behavior of kappa weighting estimators and DML estimators of the LATE. It asks which fitted estimators satisfy the relevant normalization and invariance properties. It then examines differences in effective support, concentration, sign cancellation, and covariate balance. Finally, it studies how these properties vary with the propensity specification, nuisance learner, and IV design.

The analysis makes three contributions. First, it implements five kappa estimators and two instrument-propensity estimators in the same R workflow as partially linear IV and Wald augmented inverse-probability-weighted estimators. Candidate outcome weights are recovered and checked against the reported point estimates before they enter the diagnostic analysis. Second, the classical outcome-recoding exercises are extended to the complete DML procedure. Nuisance estimation, cross-fitting, and tuning are rerun after each transformation rather than holding one fitted weight vector fixed. Third, every valid outcome-weight vector is examined with a common diagnostic sequence. The sequence covers normalization, modified effective sample size, maximum absolute weights, sign cancellation, and covariate balance. A separate positive instrument-IPW diagnostic in the Card application evaluates the design-stage propensity model without conflating it with the final outcome contrast.

The empirical analysis returns to the three applications studied by ?. This preserves comparability with their normalization results while creating a deliberate progression across IV designs. The Vietnam draft lottery of ? provides a low-dimensional benchmark with institutionally randomized assignment conditional on birth cohort. The college-proximity design of ? introduces an observational instrument, richer covariate adjustment, and two binary schooling margins with different first-stage strength. The fertility design of ? adds a large parent sample, a binary labor-supply outcome, and a logged monetary outcome. The applications differ in the source of instrument variation, the complexity of the covariate adjustment, sample size, treatment definition, and outcome scale. This variation allows the diagnostic findings to be compared across substantively different IV settings.

The results currently establish several broad patterns. The normalized kappa estimators are stable under the examined outcome translations, whereas the unnormalized estimators can change materially when only the outcome origin or monetary unit changes. The outcome-weight representation traces this sensitivity to their realized normalization properties. The DML estimators are generally stable in the low-dimensional Vietnam design, but their finite-sample weights depend on the nuisance implementation. This dependence becomes more consequential in the observational Card design, where an untuned boosted-tree specification produces extreme weights and little effective support. The diagnostics also show that similar point estimates or similar covariate balance need not imply similar normalization or sign composition. Point estimates and conventional balance plots therefore provide only a partial account of how an estimator uses the sample.
The remainder of the thesis first introduces the conditional LATE framework and defines the kappa and DML estimators. It then develops their outcome-weight representations and the links among normalization and outcome invariance. The diagnostic section defines the finite-sample measures used in the Vietnam, Card, and Angrist and Evans applications. The final sections synthesize the cross-application evidence, discuss limitations, and conclude.

2 Econometric Framework

2.1 Conditional instrumental variables, compliance types, and LATE

When treatment selection is related to potential outcomes, treated and untreated observations are not directly comparable. Instrumental-variable methods instead use treatment variation induced by the instrument. Let Y_i denote the observed outcome, $D_i \in \{0, 1\}$ the treatment, $Z_i \in \{0, 1\}$ the instrument, and X_i a vector of pretreatment covariates. Under heterogeneous treatment effects, the resulting estimand depends on whose treatment status responds to the instrument, which motivates a potential-outcomes formulation. Let $D_i(z)$ denote the treatment status that individual i would receive under instrument value $z \in \{0, 1\}$. The observed treatment is

$$D_i = Z_i D_i(1) + (1 - Z_i) D_i(0). \tag{1}$$

Before imposing exclusion, let $Y_i(z, d)$ denote the potential outcome under instrument value z and treatment status d. The exclusion restriction implies

$$Y_i(1, d) = Y_i(0, d) = Y_i(d), \qquad d \in \{0, 1\}, \tag{2}$$

so that the observed outcome satisfies

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0). \tag{3}$$

The pair $(D_i(1), D_i(0))$ partitions the population into four latent compliance types (?):

	$D_i(0) = 0$	$D_i(0) = 1$
$D_i(1) = 0$	Never-taker	Defier
$D_i(1) = 1$	Complier	Always-taker.

Always- and never-takers do not change treatment status with the instrument. Compliers respond in the direction encouraged by the instrument, whereas defiers respond in the opposite direction. Let $C_i = \mathbf{1}\{D_i(1) > D_i(0)\}$ indicate that individual i is a complier. The target parameter is the local average treatment effect (??),

$$\tau_{\text{LATE}} = E\Big[Y_i(1) - Y_i(0) \mid C_i = 1\Big]. \tag{4}$$

The parameter is defined independently of whether it can be recovered from the observed data, identification requires the conditional-IV assumptions stated below.
In many applications, the instrument is plausibly as good as randomly assigned only after conditioning on pretreatment covariates. This matters for estimation, as with heterogeneous treatment effects, additive covariate-adjusted TSLS does not generally identify a positively weighted average of complier effects. Without additional parametric restrictions, a conventional LATE interpretation requires saturated adjustment for the covariates (?). This motivates methods that incorporate the conditional instrument propensity score directly. The kappa theorem of ? provides such a route by representing moments of the latent complier population as weighted moments of observed data.

2.2 Abadie's kappa representation of complier moments

Let $p(X_i) = P(Z_i = 1 \mid X_i)$ denote the conditional probability of instrument assignment. The following conditions are Abadie's (?, Assumption 2.1) conditional version of the standard binary-IV assumptions. Their formulation here follows the notation used by ?, Assumption IV.

Assumptions (Conditional IV).

(i) Independence:

$$\Big(Y_i(0,0), Y_i(0,1), Y_i(1,0), Y_i(1,1), D_i(0), D_i(1)\Big) \perp Z_i \mid X_i.$$

(ii) Exclusion:

$$Y_i(1,d) = Y_i(0,d) \quad \text{a.s. conditional on } X_i, \quad d \in \{0,1\}.$$

(iii) First stage and overlap:

$$0 < p(X_i) < 1, \qquad P(D_i(1) = 1 \mid X_i) > P(D_i(0) = 1 \mid X_i).$$

(iv) Monotonicity:

$$P(D_i(1) \geq D_i(0) \mid X_i) = 1.$$

Condition (i) makes instrument assignment conditionally independent of the potential variables, while condition (ii) requires the instrument to affect the outcome only through treatment. Condition (iii) imposes overlap and a positive first stage in every relevant covariate stratum. Finally condition (iv) excludes defiers. Under this assumption, Abadie's (**?**, Lemma 2.1) result gives, for almost every covariate value x,

$$\begin{aligned} &E[D_i \mid Z_i = 1, X_i = x] - E[D_i \mid Z_i = 0, X_i = x] \\ &\qquad = E[D_i(1) - D_i(0) \mid X_i = x] = P(C_i = 1 \mid X_i = x) > 0. \end{aligned} \tag{5}$$

The conditional first stage therefore measures the share of compliers at $X_i = x$. Always- and never-takers contribute zero to $D_i(1) - D_i(0)$, whereas compliers contribute one. Although an individual's compliance type is not observed, Abadie's main identification result recovers complier moments from the distribution of the observed data. To state it, define

$$\kappa_{0i} = (1 - D_i)\,\frac{(1 - Z_i) - (1 - p(X_i))}{p(X_i)\big(1 - p(X_i)\big)}, \qquad \kappa_{1i} = D_i\,\frac{Z_i - p(X_i)}{p(X_i)\big(1 - p(X_i)\big)}, \tag{6}$$

$$\kappa_i = \kappa_{0i}\big(1 - p(X_i)\big) + \kappa_{1i}\,p(X_i) = 1 - \frac{D_i(1 - Z_i)}{1 - p(X_i)} - \frac{(1 - D_i)Z_i}{p(X_i)}. \tag{7}$$

Kappa representation. The following result is due to **?**, Theorem 3.1; the notation with separate functions g, g_0, and g_1 follows **?**, Lemma 2.1. Let these functions be measurable and have finite first moments. Under the Conditional IV assumption,

$$E\Big[g(Y_i, D_i, X_i) \mid D_i(1) > D_i(0)\Big] = \frac{E\Big[\kappa_i \, g(Y_i, D_i, X_i)\Big]}{P(D_i(1) > D_i(0))}, \tag{a}$$

$$E\Big[g_0(Y_i(0), X_i) \mid D_i(1) > D_i(0)\Big] = \frac{E\Big[\kappa_{0i} \, g_0(Y_i, X_i)\Big]}{P(D_i(1) > D_i(0))}, \tag{b}$$

$$E\Big[g_1(Y_i(1), X_i) \mid D_i(1) > D_i(0)\Big] = \frac{E\Big[\kappa_{1i} \, g_1(Y_i, X_i)\Big]}{P(D_i(1) > D_i(0))}. \tag{c}$$

Conditional versions of (a)-(c) hold as well. The functions κ_i, κ_{0i}, and κ_{1i} are identification weights, not probabilities, in particular, κ_i may be negative when $D_i \neq Z_i$ (**?**, p. 237). Setting $g(Y_i, D_i, X_i) = 1$ in part (a) identifies the population complier share:

$$E[\kappa_i] = P(D_i(1) > D_i(0)). \tag{8}$$

For the LATE, choose $g_0(Y_i(0), X_i) = Y_i(0)$ and $g_1(Y_i(1), X_i) = Y_i(1)$ in parts (b) and (c). This specialization, also used by **?**, Section 2, yields

$$\tau_{\text{LATE}} = \frac{E[\kappa_{1i} Y_i] - E[\kappa_{0i} Y_i]}{P(D_i(1) > D_i(0))} = \frac{E[(\kappa_{1i} - \kappa_{0i}) Y_i]}{P(D_i(1) > D_i(0))}. \tag{9}$$

Since $\kappa_{1i} - \kappa_{0i} = \frac{Z_i - p(X_i)}{p(X_i)(1-p(X_i))}$, this can also be written as

$$\tau_{\text{LATE}} = \frac{1}{P(D_i(1) > D_i(0))} \, E\left[Y_i \, \frac{Z_i - p(X_i)}{p(X_i)\Big(1 - p(X_i)\Big)}\right]. \tag{10}$$

Parts (b) and (c) with constant functions similarly imply $E[\kappa_{0i}] = E[\kappa_{1i}] = P(D_i(1) > D_i(0))$. Hence,

$$E[\kappa_i] = E[\kappa_{0i}] = E[\kappa_{1i}] = P(D_i(1) > D_i(0)). \tag{11}$$

This equality follows from Abadie's theorem. Its estimator-selection consequence is emphasized by **?**, Remark 2.2: the corresponding sample averages generally differ, even though all three estimate the same population quantity. This distinction generates the alternative kappa estimators considered next.

3 Kappa and DML-IV estimators

3.1 Kappa-based LATE estimators

Following **?**, Section 3, I compare estimators that differ in how they translate Equation (**??**) into a sample denominator. The central distinction is normalization. A normalized estimator rescales each component of an outcome contrast to have weights summing to one, whereas an unnormalized estimator uses one estimated complier share for both components. Section **??** examines the resulting implications for translation invariance and scale equivariance. The formulas initially treat $p(X_i)$ as known. In the applications it is replaced by a fitted probability $\hat{p}(X_i)$. The first estimator, $\hat{\tau}_u$, was proposed by **?** and is recommended within this class by **?**, Section 3.1. It is the ratio of separately Hájek-normalized inverse-probability-weighted contrasts for the effect of Z_i on Y_i and on D_i:

$$\hat{\tau}_u = \frac{\left(\sum_{i=1}^N \frac{Z_i}{p(X_i)}\right)^{-1} \sum_{i=1}^N \frac{Y_i Z_i}{p(X_i)} - \left(\sum_{i=1}^N \frac{1-Z_i}{1-p(X_i)}\right)^{-1} \sum_{i=1}^N \frac{Y_i(1-Z_i)}{1-p(X_i)}}{\left(\sum_{i=1}^N \frac{Z_i}{p(X_i)}\right)^{-1} \sum_{i=1}^N \frac{D_i Z_i}{p(X_i)} - \left(\sum_{i=1}^N \frac{1-Z_i}{1-p(X_i)}\right)^{-1} \sum_{i=1}^N \frac{D_i(1-Z_i)}{1-p(X_i)}}. \tag{12}$$

Thus, the inverse-probability weights are normalized within each instrument group before either contrast is formed. Appendix **??** shows the normalization algebra.
The second normalized estimator, $\hat{\tau}_{a,10}$, instead normalizes the κ_1- and κ_0-weighted outcome means separately:

$$\hat{\tau}_{a,10} = \left(\sum_{i=1}^N \kappa_{1i}\right)^{-1} \sum_{i=1}^N \kappa_{1i} Y_i - \left(\sum_{i=1}^N \kappa_{0i}\right)^{-1} \sum_{i=1}^N \kappa_{0i} Y_i. \tag{13}$$

This estimator has been considered by **?** and **?**. Its two sets of component weights each sum to one in the sample. By contrast, using one of the three sample analogues of the complier share as a common denominator gives

$$\hat{\tau}_a = \left(N^{-1} \sum_{i=1}^N \kappa_i\right)^{-1} N^{-1} \sum_{i=1}^N Y_i \frac{Z_i - p(X_i)}{p(X_i)(1-p(X_i))}, \tag{14}$$

$$\hat{\tau}_t \ (= \hat{\tau}_{a,1}) = \left(N^{-1} \sum_{i=1}^N \kappa_{1i}\right)^{-1} N^{-1} \sum_{i=1}^N Y_i \frac{Z_i - p(X_i)}{p(X_i)(1-p(X_i))}, \tag{15}$$

$$\hat{\tau}_{a,0} = \left(N^{-1} \sum_{i=1}^N \kappa_{0i}\right)^{-1} N^{-1} \sum_{i=1}^N Y_i \frac{Z_i - p(X_i)}{p(X_i)(1-p(X_i))}. \tag{16}$$

With $p(X_i)$ known or consistently estimated, these estimators inherit consistency from the same population identification argument, but both sides of their implied outcome contrast are not separately normalized. The identity $\hat{\tau}_t = \hat{\tau}_{a,1}$ and the prominence of this estimator in applied work are documented by **?**, Remark 3.1.

3.1.1 Estimating the propensity score

I estimate the instrument propensity score by the two methods considered by **?**, Section 3.5. The first fits the logit model

$$F(X_i, \alpha) = \frac{\exp(X_i'\alpha)}{1 + \exp(X_i'\alpha)}. \tag{17}$$

by maximum likelihood, producing $\hat{\alpha}^{ml}$. The second uses the covariate-balancing approach of **?**, the estimate $\hat{\alpha}^{cb}$ solves

$$N^{-1}\sum_{i=1}^{N}\frac{Z_i}{F(X_i,\hat{\alpha}^{cb})}X_i = N^{-1}\sum_{i=1}^{N}\frac{1-Z_i}{1-F(X_i,\hat{\alpha}^{cb})}X_i. \tag{18}$$

The balancing equation targets equality of reweighted covariate moments across the two instrument groups. Superscripts indicate which fitted propensity score enters an estimator. For example, $\hat{\tau}_u^{ml}$ and $\hat{\tau}_u^{cb}$ are obtained by plugging in $\hat{p}^{ml}(X_i) = F(X_i, \hat{\alpha}^{ml})$ and $\hat{p}^{cb}(X_i) = F(X_i, \hat{\alpha}^{cb})$, respectively.

With maximum-likelihood propensity scores, the alternative sample denominators generally remain distinct. With covariate balancing and a constant in X_i, however, **?**, Proposition 3.5 establish the exact finite-sample identity

$$\hat{\tau}_u^{cb} = \hat{\tau}_t^{cb} = \hat{\tau}_{a,1}^{cb} = \hat{\tau}_{a,0}^{cb} = \hat{\tau}_{a,10}^{cb}.$$

The estimator $\hat{\tau}_a^{cb}$ is not covered by this equivalence. Accordingly, the empirical sections report $\hat{\tau}_u^{cb}$ once, rather than repeating identical CBPS estimates, and report the remaining variants with maximum-likelihood propensity scores.

An important consideration for the code implementation is that no existing software implements the full set of required features in R.**?**'s `causalweight` package implements $\hat{\tau}_u$ in R, but does not support covariate balancing estimation of α and relies exclusively on the bootstrap for inference. **?**'s companion `kappalate` package supports both maximum likelihood and covariate balancing, but is implemented in Stata. Since the empirical comparison in this thesis requires the kappa estimators to sit in the same computational pipeline as the DML and outcome-weight diagnostics (all implemented in R), I implement all point estimators, both propensity-score estimation methods, and analytical standard errors directly in R (see Github for details).

3.2 DML-based IV estimators

The kappa estimators introduced above approach the conditional IV problem by constructing explicit reweighting estimators from the conditional instrument propensity score. They provide the classical benchmark for the three empirical applications considered below. However, another question that this thesis tries to solve is how these estimators compare with more flexible IV procedures that accommodate potentially complex relationships between the main variables. For this purpose, I consider IV estimators based on the double machine learning (DML) framework of **?**.

DML starts from a moment condition that identifies a low-dimensional target parameter and treats the conditional expectations entering this moment condition as nuisance functions. These nuisance functions may be estimated with flexible learners such as machine-learning methods. Because directly inserting regularized nuisance estimates into conventional moment conditions can produce substantial bias ,DML combines Neyman-orthogonal scores with cross-fitting to reduce the sensitivity of the target estimator to nuisance estimation errors.

3.2.1 The double machine learning principle and orthogonal scores

Let W_i collect the observed variables and let θ_0 denote the low-dimensional parameter of interest. The nuisance functions are collected in η_0. The parameter is characterized by a population moment condition

$$E[\psi(W_i;\theta_0,\eta_0)] = 0, \tag{19}$$

where $\psi(\cdot)$ is a score function. A direct plug-in estimator would first replace η_0 by an estimate $\hat{\eta}$ and then solve the sample analogue of (**??**). With flexible learners, however, nuisance estimates may be regularized and may adapt closely to the observations on which they were trained. The resulting nuisance-estimation error can then enter the empirical moment at first order and distort the estimate of θ_0.

DML addresses this problem first through the construction of a Neyman-orthogonal score. Consider a perturbation $\eta_0 + r(\eta - \eta_0)$ of the true nuisance functions in an admissible direction $\eta - \eta_0$. The score is Neyman orthogonal at (θ_0, η_0) if

$$\frac{\partial}{\partial r} E[\psi(W_i;\theta_0,\eta_0 + r(\eta-\eta_0))]\bigg|_{r=0} = 0. \tag{20}$$

This derivative is a Gateaux derivative with respect to the nuisance object. Condition (**??**) implies that a small local perturbation of the nuisance functions has no first-order effect on the population moment.

The distinction between first- and second-order sensitivity is central to this argument. For a non-orthogonal score, nuisance-estimation error may enter the moment linearly. If $\|\hat{\eta}-\eta_0\| = O_p(a_N)$, its contribution to the moment may therefore also be of order $O_p(a_N)$. When the score is Neyman orthogonal, the derivative in (**??**) eliminates this leading linear term. In the simplest case, the remaining contribution takes the form

$$E[\psi(W_i;\theta_0,\hat{\eta})] - E[\psi(W_i;\theta_0,\eta_0)] = O_p\Big(\|\hat{\eta}-\eta_0\|^2\Big)\,. \tag{21}$$

With several nuisance functions, second-order sensitivity more commonly appears as a product of estimation errors. For two nuisance components, for example, a remainder may be of order

$$O_p(\|\hat{\eta}_1 - \eta_{1,0}\|\|\hat{\eta}_2 - \eta_{2,0}\|)\,. \tag{22}$$

Thus, orthogonality does not eliminate nuisance-estimation error. It removes its leading linear contribution to the moment, making sufficiently small nuisance errors less consequential for the first-order distribution of the target estimator. As a familiar benchmark, if both nuisance components in (**??**) converge faster than $N^{-1/4}$, their product is $o_p(N^{-1/2})$. This can permit root-N inference for θ_0, subject to the remaining regularity conditions. The precise rate requirement depends on the score and on which nuisance-error products enter its remainder.

Orthogonality should not be interpreted as making nuisance estimation unimportant. It is a local robustness property of the score, not a guarantee against arbitrary misspecification, poor overlap, weak instruments, or severely inaccurate nuisance predictions. Furthermore, orthogonality alone does not prevent a flexible machine learning algorithm from overfitting the observations used to evaluate the score. To resolve this, DML combines the orthogonal score with cross-fitting **??**. The two devices address related but distinct problems. Orthogonality reduces the first-order sensitivity of the moment to nuisance-estimation error, whereas cross-fitting separates nuisance training from score evaluation. The precise content of θ_0, η_0, and ψ depends on the econometric model. This thesis uses two DML-based IV procedures. The first is a partially linear IV estimator, which obtains the structural coefficient from a residualized IV moment. The second is a Wald-type augmented inverse-probability-weighted estimator. Although both procedures use orthogonal scores and cross-fitted nuisance predictions, they impose different model structures and should not be assumed to identify the same causal parameter without additional restrictions. This distinction provides the organizing principle for the estimator-specific subsections that follow.

3.2.2 The partially linear IV estimator

The first DML-based IV procedure used in this thesis is the partially linear instrumental variables (PLR-IV) estimator. Following **?**, Section 4.2, consider the model

$$Y_i = D_i\theta_0 + g_0(X_i) + U_i, \qquad E[U_i \mid X_i, Z_i] = 0, \tag{23}$$

$$Z_i = m_0(X_i) + V_i, \qquad E[V_i \mid X_i] = 0. \tag{24}$$

Here, θ_0 is the low-dimensional structural coefficient of interest, whereas $g_0(\cdot)$ and $m_0(\cdot)$ are unknown nuisance functions. The covariate contribution to the outcome is left unrestricted, but the treatment enters the structural equation through the constant coefficient θ_0.

Treatment may be endogenous because equation (**??**) does not require $E[U_i \mid X_i, D_i] = 0$. Identification instead uses the conditional validity of Z_i. Equation (**??**) isolates $V_i = Z_i - m_0(X_i)$, the component of the instrument that cannot be predicted from the covariates X_i.

For estimation define the conditional means as in **?**

$$\ell_0(X_i) = E[Y_i \mid X_i], \qquad r_0(X_i) = E[D_i \mid X_i], \qquad m_0(X_i) = E[Z_i \mid X_i], \tag{25}$$

and the residualized variables

$$\widetilde{Y}_i = Y_i - \ell_0(X_i), \qquad \widetilde{D}_i = D_i - r_0(X_i), \qquad \widetilde{Z}_i = Z_i - m_0(X_i). \tag{26}$$

The corresponding partialling-out score is

$$\psi_{\mathrm{PLR-IV}}(W_i;\theta,\eta) = \Big[Y_i - \ell(X_i) - \theta\{D_i - r(X_i)\}\Big]\Big[Z_i - m(X_i)\Big], \qquad \eta = (\ell, r, m). \tag{27}$$

At the true parameter values, the score reduces to

$$\psi_{\mathrm{PLR-IV}}(W_i;\theta_0,\eta_0) = \Big(\widetilde{Y}_i - \theta_0\widetilde{D}_i\Big)\widetilde{Z}_i. \tag{28}$$

The score in (**??**) satisfies both the identifying moment condition and the Neyman-orthogonality condition introduced above.
Solving the population moment yields

$$\theta_0 = \frac{E[\widetilde{Z}_i\widetilde{Y}_i]}{E[\widetilde{Z}_i\widetilde{D}_i]}, \tag{29}$$

provided that

$$E[\widetilde{Z}_i\widetilde{D}_i] \neq 0. \tag{30}$$

Condition (**??**) is the first-stage relevance condition after partialling out X_i. Its magnitude also matters in practice as orthogonality protects against small nuisance-estimation errors, but it does not solve a weak instrument problem.
All three nuisance components in (**??**) are conditional means and can therefore be estimated directly with regression learners. Replacing the nuisance functions by cross-fitted predictions gives the sample analogue

$$\widehat{\theta}_{\mathrm{PLR-IV}} = \frac{\sum_{i=1}^{N}\{Z_i - \widehat{m}_{-k(i)}(X_i)\}\{Y_i - \widehat{\ell}_{-k(i)}(X_i)\}}{\sum_{i=1}^{N}\{Z_i - \widehat{m}_{-k(i)}(X_i)\}\{D_i - \widehat{r}_{-k(i)}(X_i)\}}, \tag{31}$$

where $k(i)$ denotes the fold containing observation i and the subscript $-k(i)$ indicates training on observations outside that fold. Thus, PLR-IV can be read as an IV regression of the residualized outcome on the residualized treatment, using the residualized instrument. The precise cross-fitting and as well the tuning procedure are described below.

Although the ratio in (**??**) resembles a Wald estimand, it should not automatically be interpreted as the LATE defined earlier. Under the maintained partially linear model, θ_0 is the constant coefficient that solves the residualized IV moment. By contrast, the LATE permits heterogeneous treatment effects and averages those effects over the complier population. Constant treatment effects are a sufficient condition for these parameters to coincide, but the equality does not generally hold under treatment-effect heterogeneity. For a binary instrument, the precise source of this distinction can be shown in the corresponding derivation is provided in Appendix **??**. Accordingly, PLR-IV is treated in the empirical analysis as a flexible residualized-IV benchmark, whereas Wald-AIPW and the normalized kappa estimators provide the more direct comparison of LATE estimators.

3.2.3 The Wald-AIPW estimator

The second DML-based IV procedure used in this thesis is the Wald augmented inverse-probability-weighted (Wald-AIPW) estimator. Unlike PLR-IV, it does not impose a partially linear outcome equation with a constant treatment coefficient. Instead, it estimates the covariate-adjusted effect of the instrument on the outcome and the corresponding effect of the instrument on treatment, and forms their ratio. The augmented Wald construction follows the doubly robust IV logic of **?**. Its implementation with orthogonal scores and cross-fitting follows the interactive IV model of **?**, Section 5.2.

For a binary instrument and binary treatment, the nuisance functions can be defined as

$$g_z(X_i) = E[Y_i \mid Z_i = z, X_i], \qquad z \in \{0,1\}, \tag{32}$$

$$r_z(X_i) = E[D_i \mid Z_i = z, X_i], \qquad z \in \{0,1\}, \tag{33}$$

$$m(X_i) = P(Z_i = 1 \mid X_i). \tag{34}$$

Here, g_z denotes the instrument-specific outcome regression, r_z the instrument-specific treatment regression, and m the conditional instrument propensity score. These nuisance functions define the covariate-adjusted reduced form and first stage as

$$\rho_Y = E[g_1(X_i) - g_0(X_i)], \tag{35}$$

$$\rho_D = E[r_1(X_i) - r_0(X_i)]. \tag{36}$$

Under exclusion, monotonicity, and a non-zero first stage, their ratio identifies the LATE

$$\tau_{\text{LATE}} = \frac{\rho_Y}{\rho_D} = \frac{E[g_1(X_i) - g_0(X_i)]}{E[r_1(X_i) - r_0(X_i)]}. \tag{37}$$

Under monotonicity, $\rho_D = E[D_i(1) - D_i(0)]$ equals the population share of compliers. Consequently, the ratio in (**??**) averages treatment effects over the complier population without imposing a common treatment effect.

A direct regression-adjustment estimator would replace the nuisance functions by fitted conditional means. Estimation errors in these regressions could, however, affect the reduced form and first stage at first order. Wald-AIPW therefore augments each regression contrast with an inverse-probability correction. Define the reduced-form pseudo-outcome

$$\Gamma_{Y,i}(\eta) = g_1(X_i) - g_0(X_i) + \frac{Z_i\{Y_i - g_1(X_i)\}}{m(X_i)} - \frac{(1-Z_i)\{Y_i - g_0(X_i)\}}{1-m(X_i)}, \tag{38}$$

and the first-stage pseudo-outcome

$$\Gamma_{D,i}(\eta) = r_1(X_i) - r_0(X_i) + \frac{Z_i\{D_i - r_1(X_i)\}}{m(X_i)} - \frac{(1-Z_i)\{D_i - r_0(X_i)\}}{1-m(X_i)}, \tag{39}$$

where $\eta = (g_0, g_1, r_0, r_1, m)$. The first terms in (**??**) and (**??**) are regression-adjustment contrasts. The remaining terms use residuals from the observed instrument branch, weighted by the inverse probability of that branch. At the true nuisance functions,

$$E[\Gamma_{Y,i}(\eta_0)] = \rho_Y, \qquad E[\Gamma_{D,i}(\eta_0)] = \rho_D. \tag{40}$$

The corresponding orthogonal score combines the two pseudo-outcomes:

$$\psi_{\text{Wald-AIPW}}(W_i; \tau, \eta) = \Gamma_{Y,i}(\eta) - \tau\Gamma_{D,i}(\eta). \tag{41}$$

This score is linear in τ and satisfies $E[\psi_{\text{Wald-AIPW}}(W_i; \tau_{\text{LATE}}, \eta_0)] = 0$. Solving the population moment therefore reproduces the adjusted Wald ratio in (**??**).

The augmentation gives the reduced-form and first-stage moments a double-robustness property. Specifically, the reduced form is consistently estimated if either the instrument propensity m or both outcome regressions (g_0, g_1) are correctly specified, while the first stage is consistently estimated if either m or both treatment regressions (r_0, r_1) are correctly specified. Hence, consistency of the ratio requires both numerator and denominator to be consistently estimated. Let $\widehat{g}_{z,-k(i)}$, $\widehat{r}_{z,-k(i)}$, and $\widehat{m}_{-k(i)}$ denote nuisance predictions obtained from models trained outside the fold containing observation i. Substituting these predictions into (**??**) and (**??**) yields the cross-fitted pseudo-outcomes $\widehat{\Gamma}_{Y,i}$ and $\widehat{\Gamma}_{D,i}$. The DML2 estimator used in the empirical analysis is

$$\widehat{\tau}_{\text{Wald-AIPW}} = \frac{N^{-1}\sum_{i=1}^{N}\widehat{\Gamma}_{Y,i}}{N^{-1}\sum_{i=1}^{N}\widehat{\Gamma}_{D,i}}. \tag{42}$$

Thus, Wald-AIPW retains the usual Wald structure, but both the adjusted reduced form and the adjusted first stage are constructed from orthogonal scores using out-of-fold nuisance predictions.

The inverse factors in (**??**) and (**??**) require the conditional instrument propensity to be bounded away from zero and one. This is not merely a technical restriction. When $\widehat{m}(X_i)$ is close to a boundary, a small number of observations can receive very large residual corrections, making the estimate sensitive to nuisance regularization and limited overlap. Orthogonality does not protect against such denominator instability. This is why the descriptive section for the three following empirical designs examines near-boundary instrument propensities. Wald-AIPW therefore provides the direct DML counterpart to the normalized kappa estimators used in this thesis. Both target the LATE under the conditional IV assumptions, but they estimate it through different sample constructions. Kappa estimators use explicit instrument-propensity weights, whereas Wald-AIPW combines instrument-specific regression adjustment with inverse-probability corrections. This difference motivates the subsequent comparison of their implied outcome weights, covariate balance, and numerical stability.

3.2.4 Cross Fitting

The DML estimators used in this thesis rely on nuisance functions that are estimated before the final treatment-effect parameter is computed. If the same observations are used both to train flexible nuisance learners and to evaluate the score, the resulting nuisance-estimation errors may be mechanically related to the score residuals. Cross-fitting mitigates this in-sample overfitting bias by evaluating each observation with nuisance predictions obtained from models that were trained without that observation (**?**).

Formally, let $I_1, \dots, I_K$ be a partition of the sample indices $\{1, \dots, N\}$. For each fold I_k, the nuisance functions are estimated using the complementary training sample $I_k^c = \{1, \dots, N\} \setminus I_k$. For an observation $i \in I_k$, the corresponding cross-fitted nuisance prediction is$\widehat{\eta}_i = \widehat{\eta}_{-k}(X_i)$, where $\widehat{\eta}_{-k}$ denotes the nuisance model trained on I_k^c. Repeating this procedure across all folds yields out-of-fold nuisance predictions for every observation. These predictions are then inserted into the orthogonal score used to estimate the target parameter.

Cross-fitting therefore separates nuisance training from score evaluation while still allowing all observations to contribute to the final estimator. Each observation is evaluated only with predictions from a model that did not use that observation for training, while it can still contribute to the training samples of the remaining folds.

In the empirical applications, I use five-fold cross-fitting. This choice provides a relatively large training sample for each nuisance model while retaining genuine out-of-fold score evaluation. Moderate fold numbers such as four or five are also recommended by **?** for practical implementation.

3.2.5 Nuisance Training

Compared to the cross-fitting procedure which determines on which observations the nuisance functions are trained and evaluated. Hyperparameter tuning determines the complexity of selected learners within the corresponding training samples. For this purpose in all tuned specifications, tuning is based on nuisance-prediction performance.

Tuning is applied in two parts of the empirical analysis. First, the honest forest smoothers used for the headline outcome-weight estimators are tuned internally by the `grf` implementation. Second, the XGBoost nuisance learners used in the DoubleML learner comparison are tuned by nested cross-validation. Linear and logistic regression are retained as fixed parametric benchmarks, while the `ranger` specifications are used as fixed off-the-shelf forest benchmarks. Thus, only XGBoost receives an explicit external tuning exercise in the learner comparison; the separate GRF-based headline estimators already use the package's native tuning procedure.

The motivation for tuning is strongest for flexible learners whose complexity can materially affect nuisance predictions. This is particularly relevant for Wald-AIPW because its score contains inverse instrument-propensity terms involving $m(X)$ and $1 - m(X)$. Predictions close to zero or one can amplify individual score contributions and make the estimate sensitive to a small number of observations. Tuning does not remove limited overlap, but it can reduce instability caused by poorly regularized nuisance models. For PLR-IV, the same inverse-propensity mechanism is absent. In this framework, tuning instead aims to improve the regression functions used to residualize the outcome, treatment, and instrument.

The headline PLR-IV and Wald-AIPW outcome-weight estimators use honest forest-based smoothers from the generalized random forest framework (**??**). Setting `tune.parameters = "all"` activates the internal GRF tuning procedure. For a regression forest, the tunable parameters are the subsample fraction, `mtry`, minimum node size, honesty fraction, honesty leaf pruning, the split-balance parameter `alpha`, and the imbalance penalty. GRF draws random hyperparameter configurations and fits a small mini-forests for each configuration. It then computes a debiased out-of-bag prediction error, smooths the relationship between the candidate parameters and these error estimates, and selects the configuration with the lowest predicted smoothed error. The final nuisance forest is grown using the selected parameters. Because the DML estimators use five-fold cross-fitting, this tuning and fitting procedure is repeated separately within the relevant outer training samples.

The XGBoost specifications are tuned through the DoubleML/`mlr3tuning` interface. Within each outer cross-fitting training sample, a random search evaluates $E = 15$ candidate configurations using three-fold inner cross-validation. The search covers the number of boosting rounds, maximum tree depth, learning rate, and minimum child weight. For nuisance component j, the selected parameter vector is

$$\widehat{\lambda}_j = \arg\min_{\lambda \in \Lambda_E} \frac{1}{K_{\text{inner}}} \sum_{k=1}^{K_{\text{inner}}} L_j(\lambda; k), \tag{43}$$

where Λ_E is the set of randomly drawn candidate configurations and $K_{\text{inner}} = 3$.
For Wald-AIPW, the outcome regressions are tuned by root mean squared error, while the binary instrument propensity and treatment regressions are tuned by log-loss. For PLR-IV, all three nuisance functions are implemented as regression learners and are therefore tuned by root mean squared error.
Tuning is performed with `tune_on_folds = TRUE`. Consequently, hyperparameters are selected separately within each outer training fold, and the observations used to evaluate the orthogonal score are excluded from both nuisance fitting and hyperparameter selection. This preserves the separation between nuisance training and score evaluation. It also implies that the tuned XGBoost specification does not have one global parameter vector; reported hyperparameters summarize the values selected across folds.

4 Connecting the Frameworks

4.1 Outcome weights and the Knaus framework

4.1.1 A common representation of the estimators

The estimators introduced in the preceding chapter approach the conditional instrumental-variables problem through markedly different sample constructions. The kappa estimators are explicit weighting estimators whose formulas depend on the conditional instrument propensity score. PLR-IV instead estimates a residualized instrumental-variables moment, whereas Wald-AIPW forms a ratio of augmented reduced-form and first-stage moments using cross-fitted nuisance estimates. These differences make a direct comparison of the estimators' defining formulas difficult. A comparison of their point estimates reveals whether they disagree in a particular sample, but it does not show how the observations in that sample are combined to produce the disagreement.
The purpose of this chapter is therefore to place the estimators in a common finite-sample diagnostic framework. The central object is a representation of a fitted estimator as a weighted sum of the observed outcomes,

$$\widehat{\tau} = \sum_{i=1}^{N} \omega_i Y_i = \boldsymbol{\omega}'\mathbf{Y}, \tag{44}$$

where $\boldsymbol{\omega} = (\omega_1, \ldots, \omega_N)'$ denotes the final outcome-weight vector. For the realized fitted procedure, each ω_i denotes the coefficient attached to observation i's outcome in the exact numerical representation of the estimate. Depending on the estimator and learner, the weight vector may itself depend on the observed outcomes and on algorithmic randomness. Equation (**??**) nevertheless converts estimators with different mathematical and computational forms into the same sample-level object.

Weighted-outcome representations are not specific to causal machine learning. They have a long tradition in weighting and matching estimators, regression, synthetic-control methods, and instrumental-variables estimation. The contribution of **?** is to provide a general framework for deriving such representations from pseudo-instrumental-variables estimators (PIVEs), to state the conditions under which the representation exists and is unique, and to obtain previously unavailable outcome weights for prominent DML and generalized-random-forest estimators. This construction provides the bridge used here between the kappa-based estimators and the DML estimators.

4.1.2 Identification weights, outcome weights, and effect weights

It is important to distinguish these outcome weights from Abadie's kappa weights. The quantities κ_i, κ_{1i}, and κ_{0i} are identification objects: under the conditional IV assumptions, they transform observed-data moments into moments of the latent complier population (**?**). The coefficients ω_i, by contrast, are properties of a particular fitted estimator in a realized sample. They are defined by the requirement that their weighted sum reproduces the numerical estimate in (**??**). Raw kappa weights may enter the construction of a kappa estimator, but they are not generally identical to its final outcome weights because the latter also incorporate the estimator's sample normalization and denominator.
Outcome weights must also be distinguished from effect weights. Outcome weights multiply observed outcomes and reproduce the fitted estimate numerically, without requiring a causal interpretation of that estimate. Effect weights instead describe how an estimator aggregates heterogeneous, and therefore generally unobserved, treatment effects across covariate values or population groups, usually as a population or expectation-based representation. The two objects answer different questions: effect weights concern the causal estimand to which an estimator gives emphasis, whereas outcome weights reveal how the realized sample is processed to obtain the reported estimate (**?**, Section 1.1).

4.1.3 The pseudo-IV representation

The PIVE framework of **?** begins from the empirical moment condition

$$\mathbb{E}_N\left[\left(\widetilde{Y}_i - \widehat{\tau}\widetilde{D}_i\right)\widetilde{Z}_i\right] = 0. \tag{45}$$

Here, $\widetilde{Y}_i$ is a pseudo-outcome, $\widetilde{D}_i$ is a pseudo-treatment, and $\widetilde{Z}_i$ is a pseudo-instrument. The prefix "pseudo" indicates that these variables need not coincide with the observed outcome, treatment, and instrument. They may contain residualization, inverse-probability adjustment, augmentation terms, or estimated nuisance functions specific to the estimator under consideration. This formulation isolates the part of the procedure that transforms the observed outcomes from the part that supplies the identifying first-stage variation.

Solving (**??**) gives

$$\hat{\tau} = \frac{\widetilde{\mathbf{Z}}'\widetilde{\mathbf{Y}}}{\widetilde{\mathbf{Z}}'\widetilde{\mathbf{D}}} = \left(\widetilde{\mathbf{Z}}'\widetilde{\mathbf{D}}\right)^{-1}\widetilde{\mathbf{Z}}'\widetilde{\mathbf{Y}}, \tag{46}$$

provided that $\widetilde{\mathbf{Z}}'\widetilde{\mathbf{D}} \neq 0$. If there exists a unique $N \times N$ transformation matrix $\mathbf{T}$ satisfying

$$\widetilde{\mathbf{Y}} = \mathbf{T}\mathbf{Y}, \tag{47}$$

then substitution into (**??**) yields

$$\hat{\tau} = \underbrace{\left(\widetilde{\mathbf{Z}}'\widetilde{\mathbf{D}}\right)^{-1}\widetilde{\mathbf{Z}}'\mathbf{T}}_{\boldsymbol{\omega}'}\,\mathbf{Y}, \tag{48}$$

and hence

$$\boxed{\boldsymbol{\omega}' = \left(\widetilde{\mathbf{Z}}'\widetilde{\mathbf{D}}\right)^{-1}\widetilde{\mathbf{Z}}'\mathbf{T}.} \tag{49}$$

Equation (**??**) is Proposition 1 of **?**. It gives a constructive two-step procedure: first, express the estimator as a PIVE; second, identify the transformation matrix $\mathbf{T}$ that maps the observed outcome vector into the pseudo-outcome. Under Knaus's condition that a unique transformation matrix $\mathbf{T}$ generates the pseudo-outcome, together with $\widetilde{\mathbf{Z}}'\widetilde{\mathbf{D}} \neq 0$, the resulting closed-form outcome-weight representation is well defined.

For the kappa estimators, the transformation matrix is diagonal. Their pseudo-outcomes are obtained by multiplying each observed Y_i directly by an observation-specific scalar constructed from Z_i, D_i, and the estimated instrument propensity score. Their outcome weights can therefore be derived algebraically without estimating an outcome-regression smoother. For PLR-IV and Wald-AIPW, the situation is different. Their pseudo-outcomes contain estimated conditional outcome functions, so a closed-form outcome-weight vector requires those predictions to admit a smoother-matrix representation. In the notation of **?**, the fitted outcome nuisance vector must be reproducible by multiplying a unique smoother matrix by $\mathbf{Y}$. Thus, the availability of DML outcome weights, unlike that of the kappa weights used here, depends on the nuisance learner and its concrete implementation.

This distinction is operational in the empirical analysis. The kappa outcome weights are computed directly from their closed-form expressions. For PLR-IV and Wald-AIPW, candidate outcome weights are extracted whenever the fitted implementation makes the required representation available. The identity $\boldsymbol{\omega}'\mathbf{Y} = \hat{\tau}$ is then verified numerically. Weight vectors that reproduce the corresponding estimate to numerical tolerance form the main diagnostic comparison. For some XGBoost implementations, an extracted vector is available but fails this identity check. Such vectors are retained only as explicitly flagged learner-sensitivity diagnostics and are not interpreted on the same footing as exact outcome-weight representations. This distinction keeps the comparison across the

Vietnam draft lottery, the Card college-proximity design, and the Angrist–Evans fertility design tied to the estimators that are actually fitted in this thesis while making the limits of the extracted representation visible.

4.1.4 Knaus's normalization classes and the empirical comparison

Once $\boldsymbol{\omega}$ has been obtained, its total mass and its masses within the treated and untreated subsamples provide a common classification of otherwise different estimators. Following **?**, Table 4, define

$$C = \boldsymbol{\omega}'\mathbf{1}_N, \qquad C_1 = \boldsymbol{\omega}'\mathbf{D}, \qquad C_0 = \boldsymbol{\omega}'(\mathbf{1}_N - \mathbf{D}). \tag{50}$$

Because $C = C_1 + C_0$, weights with zero total mass assign equal and opposite aggregate mass to treated and untreated observations. Knaus refers to weights with $C = 0$ as normalized. They are *fully normalized* when, in addition,

$$C_1 = 1 \qquad \text{and} \qquad C_0 = -1. \tag{51}$$

In that case, the treated outcomes receive aggregate weight $+1$, while the untreated outcomes receive aggregate weight -1, placing the final contrast on the conventional treated-minus-untreated scale. Individual weights within either group may nevertheless be negative. A normalized estimator need not be fully normalized: it may instead satisfy $C_1 = c$ and $C_0 = -c$ for some $c \neq 1$. Knaus calls this case scale-normalized. If $C \neq 0$, one or both sides of the comparison remain unnormalized. In particular, the untreated-unnormalized class has $C_1 = 1$ but $C_0 \neq -1$, whereas the treated-unnormalized class has $C_0 = -1$ but $C_1 \neq 1$. If neither component has its target mass, the weights are fully unnormalized (**?**, Section 4.1). The complete classification is reported in Appendix Table **??**, adapted from **?**, Table 4.

These distinctions are not merely terminological for the present comparison. Knaus derives sufficient implementation conditions determining the weight classes of PLR-IV and Wald-AIPW. Affine outcome smoothers are sufficient for zero total weight mass, while full normalization additionally requires particular alignment between the smoother matrices used for the outcome and treatment nuisance functions. Standard flexible implementations need not satisfy this stronger requirement (**?**, Sections 4.2.2 and 4.2.4). The empirical analysis therefore does not infer normalization from an estimator label. Instead, C, C_1, and C_0 are calculated for every available outcome-weight vector and learner specification.

Appendix A.4 of **?** places the kappa estimators studied by **?** within the PIVE framework and relates $\widehat{\tau}_u$ to normalized Wald-IPW. The next subsection specializes this result to the notation used in this thesis by deriving the explicit observation-level outcome weights of $\widehat{\tau}_u$. This makes transparent how the separate Hájek normalization of the instrument groups determines the total, treated, and untreated outcome-weight masses and provides the formula used in the subsequent empirical diagnostics.

4.1.5 Outcome weights of the normalized Uysal estimator

The derivation applies the PIVE construction above directly to the estimator in Equation (**??**). Let $\widehat{p}_i = \widehat{p}(X_i)$ and define the instrument-group normalizing constants

$$s_1 = \sum_{j=1}^{N} \frac{Z_j}{\widehat{p}_j}, \qquad s_0 = \sum_{j=1}^{N} \frac{1 - Z_j}{1 - \widehat{p}_j}. \tag{52}$$

The corresponding normalized instrument inverse-probability weights are

$$\lambda_{1i}^{z,n} = \frac{N Z_i / \widehat{p}_i}{s_1}, \qquad \lambda_{0i}^{z,n} = \frac{N(1 - Z_i)/(1 - \widehat{p}_i)}{s_0}. \tag{53}$$

Both vectors have total mass N, that is, $\sum_i \lambda_{1i}^{z,n} = \sum_i \lambda_{0i}^{z,n} = N$. This is precisely the separate Hájek normalization imposed on the two instrument groups in $\widehat{\tau}_u$.

Define the diagonal matrix

$$\mathbf{T}^u = \operatorname{diag}(\lambda_{11}^{z,n} - \lambda_{01}^{z,n}, \ldots, \lambda_{1N}^{z,n} - \lambda_{0N}^{z,n}). \tag{54}$$

Then the normalized reduced form and first stage in Equation (**??**) can be collected in the ratio

$$\widehat{\tau}_u = \frac{N^{-1} \mathbf{1}_N' \mathbf{T}^u \mathbf{Y}}{N^{-1} \mathbf{1}_N' \mathbf{T}^u \mathbf{D}}. \tag{55}$$

Thus, $\widehat{\tau}_u$ is a PIVE with

$$\widetilde{\mathbf{Y}} = \mathbf{T}^u \mathbf{Y}, \qquad \widetilde{\mathbf{D}} = \mathbf{T}^u \mathbf{D}, \qquad \tilde{\mathbf{Z}} = \mathbf{1}_N. \tag{56}$$

No outcome-regression smoother is required: conditional on the fitted propensity scores, $\mathbf{T}^u$ is an explicit diagonal transformation that does not depend on $\mathbf{Y}$. This applies to both the maximum-likelihood and CBPS versions of the estimator used in the empirical analysis.

For compactness, denote the normalized IPW first stage by

$$\begin{aligned} \widehat{\Delta}_D^u &:= N^{-1} \mathbf{1}_N' \mathbf{T}^u \mathbf{D} \\ &= \frac{\sum_{j=1}^N Z_j D_j / \widehat{p}_j}{s_1} - \frac{\sum_{j=1}^N (1 - Z_j) D_j / (1 - \widehat{p}_j)}{s_0}. \end{aligned} \tag{57}$$

Provided that s_1 and s_0 are finite and nonzero and that $\widehat{\Delta}^u_D \neq 0$, Equation (**??**) gives

$$\boldsymbol{\omega}^{u\prime} = \left(N^{-1}\mathbf{1}'_N\mathbf{T}^u\mathbf{D}\right)^{-1} N^{-1}\mathbf{1}'_N\mathbf{T}^u = \left(\widehat{\Delta}^u_D\right)^{-1} N^{-1}\mathbf{1}'_N\mathbf{T}^u. \tag{58}$$

Consequently, the final coefficient attached to observation i's outcome is

$$\boxed{\omega^u_i = \frac{1}{\widehat{\Delta}^u_D}\left[\frac{Z_i}{\widehat{p}_i s_1} - \frac{1-Z_i}{(1-\widehat{p}_i)s_0}\right].} \tag{59}$$

It follows immediately that $\widehat{\tau}_u = \sum_{i=1}^N \omega^u_i Y_i$. The formula also coincides with the vector returned for $\widehat{\tau}_u$ by the `kappa_outcome_weights()` function used below: each observation's signed normalized instrument weight is divided by the same estimated first stage.

Under these existence conditions, the three outcome-weight masses follow directly from the separate instrument-group normalization. First, the equal total mass of the two normalized instrument-weight vectors implies

$$\sum_{i=1}^N \omega^u_i = \frac{1}{\widehat{\Delta}^u_D}\left(\frac{1}{s_1}\sum_i \frac{Z_i}{\widehat{p}_i} - \frac{1}{s_0}\sum_i \frac{1-Z_i}{1-\widehat{p}_i}\right) = 0. \tag{60}$$

Second, the definition of $\widehat{\Delta}^u_D$ yields

$$\sum_{i=1}^N \omega^u_i D_i = 1. \tag{61}$$

Finally,

$$\sum_{i=1}^N \omega^u_i (1-D_i) = \sum_i \omega^u_i - \sum_i \omega^u_i D_i = -1. \tag{62}$$

Hence $\widehat{\tau}_u$ is fully normalized in the classification of **?**: its total outcome-weight mass is zero, while the treated and untreated masses are exactly $+1$ and -1. Importantly, this result concerns the final outcome weights, not merely the two inverse-probability weight vectors appearing in the estimator's numerator.

? proposed the normalized estimator, while **?** study its finite-sample properties. **?**, Table 3, Section 4.2.4, and Appendix A.4 identifies $\widehat{\tau}_u$ with the normalized Wald-IPW case and shows that normalization of the instrument IPW components yields fully normalized outcome weights. The derivation above specializes this general result to the notation of the kappa estimators and records the corresponding observation-level coefficient explicitly. It thereby provides the exact weight vector used in the subsequent normalization, balance, effective-sample-size, and weight-concentration diagnostics. The precise role of the separate Hájek normalization relative to $\widehat{\tau}_{a,1}$, including the special case in which covariate balancing already equalizes the two instrument-weight masses, is discussed in Appendix ??, Remark ??.

4.1.6 From derivation to implementation

The preceding algebra is implemented in a common R workflow that constructs or extracts an observation-level outcome-weight vector for each fitted estimator for which the required representation is available. For the kappa estimators, `kappa_outcome_weights(Z, D, p)` maps the observed instrument, treatment, and fitted instrument propensity scores into the five outcome-weight vectors used in the analysis. Conditional on the supplied propensity scores, these weights are evaluated from closed-form expressions. No additional numerical optimization or outcome-regression model is required at the weight-construction stage. Propensity-score estimation remains a separate preceding step. In the empirical applications, the function is evaluated using maximum-likelihood propensity scores for all five kappa estimators; the CBPS propensity scores provide the additional $\widehat{\tau}_u^{cb}$ specification.

The DML outcome weights are obtained using the `OutcomeWeights` package accompanying ?. The headline PLR-IV and Wald-AIPW specifications are fitted with `dml_with_smoother()` using five-fold cross-fitting and honest forest smoothers implemented through `grf`. Their observation-level outcome weights are then extracted with `get_outcome_weights()`. Alternative specifications are fitted as `DoubleMLPLIV` and `DoubleMLIIVM` objects using the parametric, `ranger`, and XGBoost nuisance learners. Where supported by the learner and package implementation, the stored nuisance models and predictions are passed to the `DoubleML` method of `get_outcome_weights()`.

The analysis uses the development version 0.2.0.9000 of `OutcomeWeights` at GitHub commit `0c94f940b04c`, thereby recording the implementation used for learner-specific extraction. Finally, every constructed or extracted vector is subjected to the numerical identity check described above. The companion function `check_weight_identity()` compares a constructed vector with the corresponding fitted estimate, while `check_doubleml_identity()` applies the same check to weight objects extracted from `DoubleML`. The tolerance is set to 10^{-8}. The kappa vectors satisfy the identity algebraically and reproduce their estimates to numerical precision. For the learner-based estimators, the check determines whether the extracted vector is an exact outcome-weight representation of the realized fit. Only vectors passing this check are treated as exact representations in the main comparison. Extracted vectors that fail the identity may be reported separately as learner-sensitivity evidence, but they are not interpreted on the same footing.

5 Translation and scale invariance

Normalization matters because an estimated LATE should not depend on an arbitrary change in the coding or units of the outcome. Yet, in finite samples, estimators that target the same population parameter can react differently when a constant is added to all outcomes or when wages are expressed in cents rather than dollars before taking logs. This section first explains these properties for the kappa estimators studied by **?** and then connects them to the outcome-weight representation. It then distinguishes between a fixed-weight algebraic check and a full rerun of a data-adaptive estimator. This last distinction is central to the empirical contribution of the thesis because nuisance functions and outcome weights may themselves change when a DML estimator is refitted.

5.1 Translation invariance

Translation invariance is the main outcome-transformation property examined in this thesis. It requires that an estimated treatment effect remain unchanged when the same constant is added to every observed outcome. This property is emphasized by **?**, who show that normalized and unnormalized kappa estimators can behave differently even when they target the same population LATE.

Let Y denote the vector of observed outcomes and let W collect the treatment, instrument, covariates, and all other non-outcome inputs. Following **?**, an estimator $\widehat{\tau} = \widehat{\tau}(Y, W)$ is translation invariant if

$$\widehat{\tau}(Y + k\mathbf{1}_N, W) = \widehat{\tau}(Y, W) \tag{63}$$

for every constant k.

Adding the same constant to all outcomes changes only their origin. It does not alter individual outcome differences or the underlying causal comparison. An estimator that changes after such a transformation is therefore sensitive to an arbitrary coding or centering choice.

The empirical analysis applies this idea in two settings. First, it reproduces the outcome-recoding exercises of **?** for the classical kappa estimators and the corresponding linear IV benchmarks. Second, it extends the same test to the DML estimators used in this thesis. In the latter case, the translated outcome is passed through the complete estimation procedure rather than being evaluated only with a fixed set of weights.

5.1.1 Outcome transformations in the empirical applications

The three empirical applications implement Equation (**??**) through outcome transformations that leave the economic comparison unchanged. For the binary labor-force outcome in the Angrist–Evans application, the original coding is shifted by one:

$$Y_i^{\text{shift}} = Y_i + 1, \qquad Y_i \in \{0, 1\}, \quad Y_i^{\text{shift}} \in \{1, 2\}. \tag{64}$$

Both codings preserve the distinction between working and not working. They differ only in the chosen origin of the outcome scale.

For the wage and income outcomes, the same idea appears through a change in monetary units. Let $R_i > 0$ denote the outcome in its original unit and let $Y_i = \log(R_i)$. If R_i is multiplied by a constant $a > 0$, then

$$\log(aR_i) = \log(R_i) + \log(a) = Y_i + \log(a). \tag{65}$$

A change from dollars to cents therefore adds the constant $\log(100)$ to every logged outcome. It is thus a translation-invariance test for the analyzed log outcome and, at the same time, a logarithmic scale-invariance test for the underlying monetary variable. The Vietnam and Card applications compare wages measured in dollars and cents. The Angrist–Evans application applies the corresponding unit transformations to income and also examines the binary labor-force recoding. Across all exercises, the treatment, instrument, covariates, and economic comparison are held fixed. Only the representation of the outcome is changed.

5.1.2 Translation invariance in the outcome-weight representation

The outcome-weight framework of **?** provides a direct link between normalization and translation invariance. Suppose that an estimator can be written exactly as

$$\hat{\tau}(Y, W) = \sum_{i=1}^{N} \omega_i Y_i. \tag{66}$$

If the same constant k is added to every outcome while the weights are held fixed, then

$$\hat{\tau}(Y + k\mathbf{1}_N, W) = \sum_{i=1}^{N} \omega_i (Y_i + k) \tag{67}$$

$$= \hat{\tau}(Y, W) + k \sum_{i=1}^{N} \omega_i. \tag{68}$$

Hence,

$$\hat{\tau}(Y + k\mathbf{1}_N, W) - \hat{\tau}(Y, W) = k \sum_{i=1}^{N} \omega_i. \tag{69}$$

The estimator is therefore translation invariant whenever its outcome weights sum to zero,

$$\sum_{i=1}^{N} \omega_i = 0. \tag{70}$$

This is precisely the normalization condition in the outcome-weight classification of **?**. The Knaus framework therefore gives an intuitive interpretation of the invariance result showing that a common shift in the outcome has no effect when the positive and negative outcome-weight masses cancel. For the kappa estimators considered in this thesis, the weights depend on the treatment, instrument, covariates, and estimated instrument propensity score, but not on the outcome itself. The zero-sum condition therefore directly determines whether the estimator is translation invariant. For DML estimators, however, the outcome is also used to estimate nuisance functions. Translating the outcome may therefore change the fitted nuisance models and the resulting outcome weights. For this reason, the empirical analysis evaluates translation invariance by refitting the complete DML procedure, as described in Section **??**.

5.2 Scale equivariance and logarithmic scale invariance

Translation invariance concerns additive shifts in the analyzed outcome. For positive monetary outcomes, a closely related question is whether the estimate depends on the units in which the underlying variable is measured. This issue is covered by the broader concept of scale equivariance studied by **?**.

Let $R = (R_1, \ldots, R_N)'$ denote a strictly positive outcome and define $f(R) = (g(R_1), \ldots, g(R_N))'$, where

$$g(r) = \alpha_2 r^{\alpha_1} - \alpha_3.$$

An estimator is scale equivariant if

$$\hat{\tau}(f(aR), W) = a^{\alpha_1} \hat{\tau}(f(R), W) \tag{71}$$

for every $a > 0$.

The empirical analysis in this thesis does not examine this full class of transformations. It focuses on its natural-logarithm special case. Setting $\alpha_2 = \alpha_3 = 1/\alpha_1$ and taking the limit $\alpha_1 \to 0$ yields $g(r) = \log(r)$. The corresponding requirement is

$$\hat{\tau}(\log(aR), W) = \hat{\tau}(\log(R), W)\,. \tag{72}$$

This property is referred to as scale invariance with respect to the natural logarithm. It requires that an estimate based on logged wages or income remain unchanged when the underlying monetary variable is expressed in a different unit. For example, replacing dollars by cents multiplies R_i by 100, but

$$\log(100R_i) = \log(R_i) + \log(100).$$

The dollars-to-cents comparison is therefore a logarithmic scale-invariance test for the underlying wage or income variable. At the same time, it is a translation-invariance test for the logged outcome used in estimation.

The Vietnam and Card applications apply this comparison to wages, while the Angrist–Evans application considers several alternative units for income. These exercises reproduce the classical kappa comparison and extend the same test to the DML estimators used in this thesis.

5.3 Empirical assessment of estimator invariance

The preceding results establish a direct connection between normalization and translation invariance. For the kappa estimators, this connection can be evaluated directly because their fitted weights depend on the instrument, treatment, covariates, and estimated instrument propensity score, but not on the outcome. Re-estimating a kappa estimator after translating the outcome therefore provides a transparent finite-sample illustration of the invariance results derived by **?**.

In the following the empirical applications first reproduce these classical outcome-transformation comparisons. In the Vietnam and Card settings, the relevant specifications are applied to the original samples used in the corresponding replication exercises. In the Angrist–Evans setting, the same comparison is conducted on the diagnostic subsamples used for the outcome-weight and DML analysis. These subsamples preserve the main features of the empirical IV design but are not intended to reproduce the published full-sample estimates numerically. Their purpose is to examine whether the same finite-sample invariance mechanism remains visible within the samples on which the complete diagnostic pipeline can be implemented.

The extension to DML requires an additional step. A DML estimate depends on fitted nuisance functions and, depending on the learner, on cross-fitting, randomization, model selection, and hyperparameter tuning. When the outcome is translated, these components may be estimated differently. Translation invariance must therefore be assessed for the entire estimation procedure rather than only for one realized outcome-weight vector.

Let $\mathcal{A}\Big(Y, W; \mathcal{L}, \mathcal{T}, \mathcal{F}, s\Big)$ denote a particular estimator implementation, where $\mathcal{L}$ is the learner specification, $\mathcal{T}$ the tuning procedure, $\mathcal{F}$ the cross-fitting scheme, and s the random seed. The original and translated estimates are defined as

$$\widehat{\tau}_{\text{orig}} = \mathcal{A}\Big(Y, W; \mathcal{L}, \mathcal{T}, \mathcal{F}, s\Big), \tag{73}$$

$$\widehat{\tau}_{\text{shift}} = \mathcal{A}\Big(Y + k\mathbf{1}_N, W; \mathcal{L}, \mathcal{T}, \mathcal{F}, s\Big). \tag{74}$$

The complete-pipeline invariance statistic is

$$\Delta_{\text{rerun}} = \widehat{\tau}_{\text{shift}} - \widehat{\tau}_{\text{orig}}. \tag{75}$$

An implementation is invariant to the examined translation when Δ_{rerun} is zero.

The comparison is designed to isolate the outcome transformation as closely as possible. The original and translated fits use the same analysis sample, estimator, learner class, number of cross-fitting folds, tuning rule, tuning search space, tuning budget, loss function, and random seed. The seed is reset before each fit. The current implementation regenerates the folds from the same seed and call sequence rather than storing and manually reinserting a common fold object. For data-adaptive learners, the fitted nuisance functions and selected hyperparameters are re-estimated after translating the outcome.

The rerun check is applied to the normalized and unnormalized kappa estimators, the linear IV benchmark, PLR-IV, and Wald-AIPW. For the DML estimators, the analysis includes the tuned honest-forest implementation, parametric nuisance models, `ranger`, and XGBoost where the required tuning results are available. The XGBoost nuisance models for PLR-IV and Wald-AIPW are tuned separately because the two estimators require different nuisance functions and optimize different prediction problems. For the translated outcome, the nested tuning procedure is rerun under the same search design used for the original outcome.

The resulting analysis uses the classical kappa results as a benchmark and examines whether the same invariance is preserved across increasingly data-adaptive IV estimation procedures.

6 Outcome-weight diagnostics

The preceding sections establish the outcome-weight representation and its normalization classes. The diagnostic analysis uses these results to describe the finite-sample comparison behind each estimate. It starts from the numerical identity

$$\hat{\tau} = \sum_{i=1}^{N} \omega_i Y_i. \tag{76}$$

The identity is checked to numerical tolerance for every fitted estimator. In this thesis the following diagnostic workflow examines four properties of every exact weight vector. The weight masses describe the normalization of the fitted contrast. The distribution of absolute weight magnitudes describes effective support and concentration. The sign composition records the degree of cancellation within each treatment component. The covariate diagnostics characterize the observed comparison generated by the final weights. These properties are reported together because they describe different features of the same estimator.

6.1 Normalization in the empirical diagnostics

The normalization classes and their implications for translation invariance have already been derived in Section **??** and Section **??**. The empirical tables implement that classification directly. They report the total mass C and the treated mass C_1 from Equation (**??**). The untreated mass is displayed as $-C_0$. This sign change places both treatment components on the same positive orientation. A fully normalized contrast therefore appears as $C = 0$ and $C_1 = -C_0 = 1$.

This presentation serves two purposes. It verifies the normalization properties predicted by the estimator formulas. It also makes departures from unit-mass normalization comparable across kappa and DML estimators. The calculation is applied to the realized weight vector rather than inferred from the estimator name. This follows the implementation emphasis of **?**, since learner and smoother choices can determine the normalization properties of DML outcome weights.

6.2 Effective support and concentration

An estimator may reproduce a normalized contrast while assigning most of its absolute weight mass to a small part of the sample. I measure this concentration with the modified effective sample size proposed by **?**, Section 5.4. For treatment state $d \in \{0, 1\}$, it is

$$ESS_{\text{mod},d} = \frac{\left(\sum_i \mathbf{1}\{D_i = d\}|\omega_i|\right)^2}{\sum_i \mathbf{1}\{D_i = d\}\omega_i^2}. \tag{77}$$

The absolute values preserve information about weight magnitudes when both signs are present. The measure lies between one and the number of observations in the respective treatment group. It coincides with the conventional Kish effective sample size when the weights are nonnegative and normalized (**?**).

?, Section 5.4 recommend computing the measure separately for treated and control observations for both the URI and MRI regression weights studied in their paper. I apply the same group-specific diagnostic principle to the signed IV outcome weights. In this setting, the measure is interpreted as a scale-invariant summary of absolute-weight concentration.

The empirical tables report both the group-specific level and its share of the available treatment arm. Let

$$N_d = \sum_{i=1}^{N} \mathbf{1}\{D_i = d\}, \qquad \rho_d = \frac{ESS_{\mathrm{mod},d}}{N_d}. \tag{78}$$

The value ρ_d is bounded by zero and one. A value close to one indicates that the absolute weight magnitudes are distributed almost uniformly within treatment arm d. A smaller value indicates that the same component relies on a narrower part of the observed arm. The overall modified effective sample size is retained as a supplementary summary. The group-specific values are the main diagnostic because concentration in one treatment component can be hidden by the combined measure. I also report $\max_i |\omega_i|$. It records the largest observation-level coefficient in the fitted outcome contrast. The modified effective sample size describes the full distribution of magnitudes, while the maximum records its most pronounced tail.

6.3 Sign composition and cancellation

The treated and untreated components enter an outcome contrast with opposite aggregate signs. Following the orientation used for the descriptive weight analysis in **?**, Section 5, I define

$$a_i = \omega_i(2D_i - 1). \tag{79}$$

The treated weights remain unchanged and the untreated weights are multiplied by minus one. Positive values of a_i therefore follow the conventional orientation in both treatment arms.

The empirical analysis summarizes the remaining negative component by its share of absolute weight mass

$$R_d^- = \frac{\sum_i \mathbf{1}\{D_i = d\}|a_i|\mathbf{1}\{a_i < 0\}}{\sum_i \mathbf{1}\{D_i = d\}|a_i|}. \tag{80}$$

This statistic gives greater importance to a large opposite-oriented weight than to a numerically negligible one. It therefore differs from an unweighted count of negative observations. The statistic measures cancellation. Let P_d denote positive oriented mass and let N_d^- denote the absolute mass of negative oriented weights. The signed group mass is $P_d - N_d^-$, while total absolute mass is $P_d + N_d^-$. Hence $R_d^- = N_d^-/(P_d + N_d^-)$. Values close to one half indicate that sizeable positive and negative components offset each other. Values closer to zero indicate that most absolute mass follows one orientation. The cancellation statistic complements the modified effective sample size. The latter depends on absolute magnitudes and is invariant to their signs. A weight vector can therefore have a large modified effective sample size and still contain substantial cancellation. Reading $ESS_{\text{mod},d}$, ρ_d, R_d^-, and $\max_i |\omega_i|$ together separates diffuse weighting from sign offsetting and individual weight dominance.

6.4 Covariate balance

The concentration diagnostics describe how the sample is weighted. Covariate balance describes the observed comparison produced by those weights. This is useful when two estimators yield similar point estimates but construct their fitted contrasts from different regions of the covariate distribution. It also permits the kappa and DML estimators to be assessed with the same sample-level object. For covariate X_j and treatment state d, the oriented weighted mean is

$$\bar{X}_{dj}^a = \frac{\sum_i \mathbf{1}\{D_i = d\} a_i X_{ij}}{\sum_i \mathbf{1}\{D_i = d\} a_i}, \qquad d \in \{0, 1\}, \tag{81}$$

provided that the denominator is nonzero. This normalization makes the comparison invariant to a common rescaling within either treatment component. The implemented balance measure distinguishes continuous and binary covariates. Let s_{dj}^2 denote the unweighted variance of a continuous covariate in treatment group d. Define the pooled scale as $s_{j,\text{pool}} = [(s_{1j}^2 + s_{0j}^2)/2]^{1/2}$. The same denominator is used for the unadjusted and weighted comparisons. The reported balance statistic is

$$B_j(a) = \begin{cases} \dfrac{\left|\bar{X}_{1j}^a - \bar{X}_{0j}^a\right|}{s_{j,\text{pool}}}, & X_j \text{ continuous,} \\ \left|\bar{X}_{1j}^a - \bar{X}_{0j}^a\right|, & X_j \text{ binary.} \end{cases} \tag{82}$$

Continuous covariates are therefore expressed in units of the pooled standard deviation. Binary covariates remain on their raw zero–one scale. This avoids inflating the reported imbalance of rare binary indicators through standardization. The weighted means require careful interpretation when some a_i are negative. A binary weighted mean is then an algebraic mean of an indicator. It need not lie between zero and one. The balance statistic remains a valid moment comparison, but it does not describe two nonnegative weighted proportions. The opposite-sign mass in Equation (**??**) shows how much cancellation accompanies the reported balance.

6.4.1 Implementation and visualization

The balance statistics are displayed with Love plots constructed using the `cobalt` package (?). This follows the application of outcome weights to canonical covariate-balance plots in ?, Section 5. Uniform weights provide the unadjusted treatment comparison. The estimator-specific a_i provide the adjusted comparison. Continuous and binary covariates are marked separately. A reference line at 0.1 corresponds to one tenth of a pooled standard deviation for continuous covariates and to a raw difference of 0.10 for binary covariates.

6.4.2 Instrument-propensity diagnostics

The Card application also evaluates the fitted instrument propensity score before examining the final outcome contrast. The two instrument groups receive positive inverse-probability weights proportional to $Z_i/\widehat{p}_i$ and $(1 - Z_i)/(1 - \widehat{p}_i)$. Each group is normalized separately. Covariate balance then records how closely the weighted $Z = 1$ and $Z = 0$ samples resemble each other. Conventional Kish effective sample size is reported within each instrument group because these design weights are nonnegative.

This diagnostic concerns the construction of the instrument propensity model. The outcome-weight diagnostics concern the final comparison between $D = 1$ and $D = 0$ that reproduces the reported estimate. Their joint use connects the estimated design stage to the geometry of the fitted estimator without treating the two comparisons as equivalent.

The empirical applications now apply this diagnostic sequence to three distinct IV designs. Each application first establishes the relevant data and point estimates. It then examines normalization, effective support, sign composition, and covariate balance for the exact outcome-weight representations. The Vietnam draft lottery provides the first application and serves as the low-dimensional benchmark for the subsequent Card and Angrist and Evans comparisons.

7 Empirical Application: Military Service and Wages

The first application revisits **?**, and studies the effect of Vietnam-era military service on later earnings. Veteran status is not randomly assigned. Men who served may differ from nonveterans in their health, education, and especially in their willingness to enlist. A direct comparison of the two groups would therefore combine the effect of military service with selection into the military. In order to overcome this issue this design uses draft eligibility as an instrument for military service.

7.1 Data and Design

During the Vietnam-era draft lotteries, random sequence numbers were assigned to dates of birth. For each relevant birth cohort, an induction ceiling determined whether an individual was classified as draft eligible. Define

$$Z_i = \mathbf{1}\Big\{RSN_i \leq c_{b(i)}\Big\}\,, \tag{83}$$

where RSN_i is individual i's random sequence number, $b(i)$ denotes his birth cohort, and $c_{b(i)}$ is the corresponding eligibility ceiling. Appendix Table **??** reports the cohort-specific ceilings used in the sample. Because the eligibility ceilings differed across cohorts, the probability of draft eligibility also varied across cohorts:

$$P(Z_i = 1 \mid b(i)) = P\Big(RSN_i \leq c_{b(i)} \mid b(i)\Big)\,. \tag{84}$$

Draft eligibility is therefore treated as as-good-as randomly assigned conditional on the cohort structure. Following **?**, the empirical specifications use age as the conditioning variable that captures this structure.

The analysis uses the **?** subsample of the 1984 Survey of Income and Program Participation used by **?**. The final sample contains $N = 3{,}027$ white men born between 1950 and 1953 with observed draft-eligibility, veteran-status, and wage information. In the notation of Section **??**, Z_i denotes draft eligibility, D_i veteran status, and Y_i the logarithm of hourly wages.

Compliance with draft eligibility was incomplete. Some eligible men did not serve, while some ineligible men entered the military voluntarily. Nevertheless, eligibility increased the probability of military service and thereby generated the first-stage variation used in the analysis. In the pooled sample, 45.6% of men were draft eligible. Veteran rates were 40.4% among eligible men and 26.5% among ineligible men, corresponding to an unadjusted first-stage difference of 13.9 percentage points. Appendix Table **??** reports the main design diagnostics. After conditioning on age, the estimated first stage is approximately nine percentage points across the three specifications. The squared HC1-robust

first-stage t-statistics range from 26.3 to 30.1. The values do not indicate a weak first stage in this sample, but still should be understood as descriptive strength diagnostics.
The three age specifications differ in their flexibility. The linear model provides a parsimonious benchmark, while the cubic model allows for a smooth nonlinear relationship between age and draft eligibility. The saturated specification assigns a separate eligibility probability to each observed age category and therefore reflects the discrete cohort structure of the lottery. It is used as the main flexible specification below, with the linear and cubic models providing sensitivity checks.
Estimated instrument propensity scores remain well inside the unit interval. Across the three specifications, fitted probabilities range from approximately 0.14 to 0.63, and none falls below 0.05 or above 0.95.Instability arising solely from extreme instrument propensities is therefore unlikely in this application. Appendix Figure **??** shows the fitted propensity-score distributions. Relative to the linear specification, the cubic and saturated models produce more clustered fitted values, with the saturated model generating distinct mass points by age group.

7.2 Point Estimates across Age Specifications

Before examining translation invariance, this subsection examines how the estimated effect of military service changes across the linear, cubic, and saturated age specifications. The comparison first considers the 2SLS and kappa estimates reported in the replication of **?**. It then adds PLR-IV and Wald-AIPW estimates based on honest-forest nuisance functions. Appendix Table **??** reports all estimates using log hourly wages measured in dollars.
Under the linear age specification, 2SLS and the three normalised kappa estimators produce very similar point estimates. The 2SLS estimate is 0.233, while the normalised kappa estimates range from 0.227 to 0.234. By contrast, the three unnormalised kappa estimates lie between 0.014 and 0.016. The disagreement under linear age therefore concerns the unnormalised kappa estimators. It is not a general difference between 2SLS and weighting estimators, as 2SLS is close to all three normalised kappa estimates.
The cubic specification produces a different pattern. The normalised kappa estimates are lowest and range from 0.202 to 0.210. The 2SLS estimate is 0.227, while the honest-forest estimates equal 0.243 for PLR-IV and 0.229 for Wald-AIPW. The unnormalised kappa estimates are higher and range from 0.302 to 0.317. The cubic specification therefore separates the estimators into three groups. The normalised kappa estimates lie at the lower end, 2SLS and the honest-forest DML estimates occupy the middle of the range, and the unnormalised kappa estimates lie at the upper end. This is the age specification under which the choice of estimator has the clearest effect on the reported point estimate.
The estimates become much more similar under the saturated age specification. All six

kappa estimators equal 0.241 in this sample. The PLR-IV and Wald-AIPW estimates are 0.242 and 0.244, respectively, while the 2SLS estimate is 0.254. The complete range is therefore only 0.013. Among the three age specifications, the saturated specification produces the greatest numerical agreement across estimators. This agreement should be interpreted in light of the saturated age-cell structure, under which the six kappa estimators coincide in this sample.

The honest forest is only one possible choice for estimating the DML nuisance functions. Appendix Table **??** therefore compares it with regression models, random forests estimated through `ranger`, and XGBoost. All specifications use five-fold cross-fitting.

Under cubic age, the PLR-IV estimates range from 0.239 with regression nuisances to 0.270 with the random forest. The honest-forest estimate of 0.243 is close to the regression estimate, while the random-forest and XGBoost estimates are higher by 0.027 and 0.022, respectively. The cubic Wald-AIPW estimates range from 0.218 to 0.246. Relative to the honest-forest estimate of 0.229, the largest difference is 0.017. The nuisance learner has a somewhat larger effect on PLR-IV under saturated age. The honest-forest estimate is 0.242, compared with 0.270 for the regression specification, 0.283 for the random forest, and 0.281 for XGBoost. The corresponding Wald-AIPW estimates are more stable. They range from 0.244 with the honest forest to 0.261 with the random forest. Overall, the nuisance learner changes the PLR-IV point estimates more than the Wald-AIPW estimates, particularly under saturated age. Nevertheless, all DML estimates remain positive and within a relatively narrow overall range from 0.218 to 0.283.

The honest-forest and XGBoost results reported here use tuned nuisance models. A separate tuning exercise shows that tuning changes the honest-forest estimates by at most 0.010 and the XGBoost estimates by at most 0.011. Tuning therefore does not materially affect the conclusions of the Vietnam application. It is not examined further in this subsection and receives more attention in the later applications where its effect is more substantial.

7.3 Translation Invariance under Complete Re-estimation

Measuring wages in cents instead of dollars adds log(100) to the log-wage outcome without changing the underlying economic quantity. This provides a direct test of translation invariance. The classical results reproduce the central pattern documented by **?**. The 2SLS and normalised kappa estimates are unchanged when wages are recoded from dollars to cents. Across the three age specifications, these estimates range from 0.202 to 0.254. The unnormalised kappa estimators are considerably more sensitive. Under linear age, for example, $\hat{\tau}_a$ changes from 0.015 with wages in dollars to -0.429 with wages in cents. Under cubic age, the three unnormalised estimates increase by between 0.213 and 0.224 log points after the translation. Under saturated age indicators, all six kappa estimators equal 0.241 under both outcome units. This numerical agreement is a feature of the saturated age-cell specification in this sample. The classical estimates match Table 2 of **?** at the reported precision, apart from the cubic CBPS estimate $\hat{\tau}_u^{cb}$. The tightly solved CBPS implementation used in this thesis gives an estimate of 0.210 with a standard error of 0.233. The published estimate and standard error are 0.208 and 0.232, respectively. Appendix Section **??** shows that this small difference is caused by the numerical solution of the CBPS balancing conditions rather than by the sample or estimator definition.

For the DML comparison, each complete estimation procedure is fitted once to log wages in dollars and once to the translated outcome. The same sample, cross-fitting design, number of folds, and random seed are used in both runs. All outcome nuisance functions are re-estimated. Honest-forest tuning is repeated internally, while the XGBoost specifications repeat nested tuning within each outer training fold. The DML exercise is restricted to cubic and saturated age.

The honest-forest, regression, and `ranger` estimates are numerically unchanged after complete re-estimation. The OLS and `ranger` estimates are identical across the two outcome codings at the reported precision. The same holds for three of the four honest-forest estimates. The only nonzero honest-forest difference is 0.000018 for saturated Wald-AIPW. This lies below the numerical threshold of 10^{-4} used in the notebook and is treated as numerical rather than substantive variation.

These findings are consistent with the outcome-weight results in **?**, Condition 3 and Table 5. When the outcome nuisance is estimated by an affine smoother, its smoother-matrix rows sum to one. This is sufficient for the fitted PLR-IV and Wald-AIPW outcome weights to sum to zero. The results in Table **??** show that the corresponding implemented procedures also remain numerically invariant when the complete estimation pipeline is rerun.

The tuned XGBoost specifications produce small but nonzero changes in all four cases. The largest change is 0.002917 for cubic PLR-IV. Saturated Wald-AIPW changes by -0.001528, while the remaining changes are below 0.001 in absolute value. These differences exceed the numerical threshold used in the notebook, so the XGBoost procedures are not classified as numerically invariant. This result is also consistent with the qualification in **?**. Knaus treats XGBoost as a non-affine smoother in his empirical illustration and notes that boosted trees can be an exception to the affine-smoother condition. The normalization result available for affine smoothers therefore does not automatically extend to the implemented XGBoost procedures. In addition, nested tuning may select a slightly different fitted nuisance model after the outcome is translated. The XGBoost changes remain small relative to the point estimates. The Vietnam application is therefore seemingly stable across the two outcome codings, even though exact numerical invariance is not obtained for the tuned boosted-tree specifications. The next subsection examines the realised outcome weights and separates numerical reproduction of the estimates from their normalization and concentration properties.

Table 1: Translation invariance under complete re-estimation

Estimator	Cubic age			Saturated age		
	Dollars	Cents	Change	Dollars	Cents	Change
Panel A: Kappa estimators						
$\hat{\tau}_u^{cb}$	0.209995	0.209995	0.000000	0.240810	0.240810	0.000000
$\hat{\tau}_u^{ml}$	0.202483	0.202483	0.000000	0.240810	0.240810	0.000000
$\hat{\tau}_{a,10}$	0.204434	0.204434	0.000000	0.240810	0.240810	0.000000
$\hat{\tau}_a$	0.314421	0.536590	0.222169	0.240810	0.240810	0.000000
$\hat{\tau}_t$	0.302027	0.515439	0.213412	0.240810	0.240810	0.000000
$\hat{\tau}_{a,0}$	0.316704	0.540486	0.223782	0.240810	0.240810	0.000000
Panel B: Tuned honest forest						
PLR-IV	0.242913	0.242913	0.000000	0.241573	0.241573	0.000000
Wald-AIPW	0.229085	0.229085	0.000000	0.244280	0.244297	0.000018
Panel C: Alternative nuisance learners						
PLR-IV, OLS	0.238774	0.238774	0.000000	0.270119	0.270119	0.000000
Wald-AIPW, OLS/logit	0.217826	0.217826	0.000000	0.246814	0.246814	0.000000
PLR-IV, `ranger`	0.269686	0.269686	0.000000	0.283419	0.283419	0.000000
Wald-AIPW, `ranger`	0.246134	0.246134	0.000000	0.260978	0.260978	0.000000
PLR-IV, tuned XGBoost	0.264828	0.267745	0.002917	0.280918	0.280400	−0.000518
Wald-AIPW, tuned XGBoost	0.238268	0.237958	−0.000310	0.253258	0.251730	−0.001528

Notes: Log wages in cents equal log wages in dollars plus log(100). “Cents” reports the estimate after the complete procedure has been re-estimated using the translated outcome.

7.4 Outcome-Weight Diagnostics

This section examines the realised outcome weights behind the Vietnam estimates. The main discussion focuses on cubic age adjustment because this specification produces the clearest differences across the alternative weighting constructions. The complete cubic and saturated results are reported in Appendix Table **??**.

The weight sums and group masses provide the normalisation classification described in **?**, Table 4. The three normalised kappa estimators satisfy all restrictions for full normalisation. Their total weights sum to zero, while the treated and sign-oriented control masses both equal one. The remaining kappa estimators belong to three different classes. For $\hat{\tau}_a$, neither group mass equals one, so the estimator is fully unnormalised. For $\hat{\tau}_t$, the treated mass equals one but the control mass is only 0.954, which makes it untreated-unnormalised. For $\hat{\tau}_{a,0}$, the control mass equals one but the treated mass is 1.049, which makes it treated-unnormalised. These realised weights reproduce the classification derived for the kappa estimators in **?**, Appendix A.4.

The DML results show why the nuisance implementation matters. The honest-forest PLR-IV and Wald-AIPW weights sum to zero, while the treated and control masses are equal within each estimator. They are therefore scale-normalised. Their common masses of 0.9942 and 0.9998 are close to one but do not satisfy full normalisation exactly. This pattern is consistent with **?**, Table 5. Affine outcome smoothers satisfy Condition 3 and therefore produce normalised weights. Full normalisation additionally requires Condi-

Table 2: Selected outcome-weight diagnostics under cubic age adjustment

Estimator	$\sum_i \omega_i$	Mass T/C	Opposite-sign mass T/C (%)	$\text{ESS}_{\text{mod}}/N$ T/C
DML estimators				
PLR-IV, honest forest	0.0000000	0.9942/0.9942	43.5/46.6	0.963/0.921
Wald-AIPW, honest forest	0.0000000	0.9998/0.9998	43.4/46.7	0.952/0.910
PLR-IV, OLS	0.0000000	1.0000/1.0000	43.6/46.7	0.966/0.924
Wald-AIPW, OLS/logit	0.0000000	0.9915/0.9915	43.5/46.8	0.955/0.908
Kappa estimators				
$\hat{\tau}_u^{cb}$	0.0000000	1.0000/1.0000	43.5/46.8	0.958/0.908
$\hat{\tau}_u^{ml}$	0.0000000	1.0000/1.0000	43.6/46.8	0.954/0.899
$\hat{\tau}_{a,10}$	0.0000000	1.0000/1.0000	43.5/46.9	0.954/0.898
$\hat{\tau}_a$	0.0482434	1.0410/0.9928	43.5/46.9	0.954/0.898
$\hat{\tau}_t$	0.0463418	1.0000/0.9537	43.5/46.9	0.954/0.898
$\hat{\tau}_{a,0}$	0.0485937	1.0486/1.0000	43.5/46.9	0.954/0.898

Notes: T/C denotes the observed treated and control groups. Mass T/C reports $\sum_{i:D_i=1} \omega_i$ and $-\sum_{i:D_i=0} \omega_i$, so both components follow the same positive orientation. Opposite-sign mass is the percentage of absolute weight mass that does not follow this conventional orientation. The modified ESS is calculated separately within the observed treated and control groups and divided by the corresponding group size. The complete diagnostics, including the `ranger` and reconstructed XGBoost results, are reported in Appendix Tables **??** and **??**.

tion 5a for PLR-IV or Condition 5b for Wald-AIPW, which is generally not satisfied when different nuisance models are used.

The OLS implementation of PLR-IV displays the fully normalised pattern, with a total weight sum of zero and both group masses equal to one. This is consistent with the same linear projection structure being used for the relevant nuisance functions. Wald-AIPW combines linear outcome regressions with logistic models for the binary nuisance functions. Its total weights sum to zero, but both group masses equal approximately 0.9915. It is therefore scale-normalised rather than fully normalised.

The modified ESS ratios describe a different property of the weights. Following the definition in Section **??**, values close to one indicate that the absolute weight magnitudes are distributed relatively evenly within the corresponding treatment group. Across the estimators in Table **??**, the ratios range from 0.952 to 0.966 among veterans and from 0.898 to 0.924 among nonveterans. The lower values for the control group indicate somewhat greater concentration among nonveterans, although none of the reported ratios suggests that the estimates rely on a very small part of either group.

PLR-IV has slightly higher modified ESS ratios than Wald-AIPW for both nuisance implementations. With honest forests, the ratios are 0.963 and 0.921 for PLR-IV, compared with 0.952 and 0.910 for Wald-AIPW. The same ordering appears for the linear and logistic nuisance models.

The kappa estimators provide another useful comparison. Their modified ESS ratios remain very similar even though their normalisation classes differ. In particular, $\hat{\tau}_{a,10}$, $\hat{\tau}_a$,

$\hat{\tau}_t$, and $\hat{\tau}_{a,0}$ have almost identical concentration diagnostics. The differences between their realised weighting structures therefore arise mainly from their aggregate weight masses.
The opposite-sign mass reveals substantial cancellation within both treatment groups. Approximately 43.4% to 43.6% of the absolute treated weight mass and 46.6% to 46.9% of the absolute control weight mass have the opposite sign from the conventional orientation. The control weights are therefore slightly closer to an even split between the two signs. However, the values are remarkably similar across the DML and kappa estimators. The cancellation is consequently a common feature of the weighting structure in this application rather than a problem generated by one particular estimator or learner.
Substantial opposite-sign mass does not by itself imply that an estimator is invalid, nor does it provide evidence for or against individual-level monotonicity. It shows that positive and negative components offset within the realised outcome contrast. This result can coexist with high modified ESS ratios because the modified ESS depends on absolute weight magnitudes and not on their signs. Taken together, the two diagnostics show that the Vietnam weights are broadly distributed across the observed sample but involve considerable internal cancellation.

7.5 Covariate Balance

Following the previous covariate diagnostics, Appendix Figures **??** and **??** compare covariate balance before and after weighting. Only exact or algebraic outcome-weight representations are included. The cubic specification reports standardised differences for age, age squared, and age cubed, while the saturated specification reports raw mean differences for binary age indicators. Their magnitudes should consequently be compared within, rather than across, specifications.
Under cubic age adjustment, the unadjusted standardised differences range from approximately 0.52 to 0.56. The CBPS-based estimator $\hat{\tau}_u^{cb}$ balances all three age terms up to numerical precision, which reflects the balancing restrictions imposed by CBPS. The maximum-likelihood estimator $\hat{\tau}_u^{ml}$ reduces the differences substantially but retains a maximum value of 0.109, slightly above the 0.1 reference line. The estimate $\hat{\tau}_{a,10}$ also remains close to this line. Among the DML estimators, tuned-forest PLR-IV has a maximum adjusted difference of 0.099, whereas honest-forest Wald-AIPW and the eligible linear and `ranger` specifications balance the included terms more closely. These results show that full normalisation does not imply exact covariate balance, since the fully normalised kappa estimators differ according to how their instrument propensity scores are estimated.
Under saturated age adjustment, the remaining differences are small for all included estimators. The six kappa estimators balance the age indicators up to numerical precision because their realised weights coincide in this specification. Honest-forest PLR-IV is also close to exact balance, while the maximum differences for honest-forest Wald-AIPW

and the `ranger` specifications remain below approximately 0.03. The irregular patterns across individual age indicators therefore reflect small differences on a narrow plotting scale rather than a general balance problem.

Overall, the balance diagnostics provide no evidence of substantial remaining imbalance in the Vietnam application. The main differences appear under cubic adjustment, where CBPS achieves exact balance while the MLE-based kappa weights and tuned-forest PLR-IV retain differences close to the 0.1 reference value.

7.6 Summary of the Vietnam Application

The Vietnam application links the reproduction pipeline to the DML and outcome-weight analysis. The cents and dollars comparison reproduces the main pattern documented by **?**. The normalised kappa estimators are unchanged after the outcome translation, while the incompletely normalised estimators change under cubic age adjustment. Under saturated age indicators, all six kappa estimators coincide in this sample.

The DML estimates are also stable across the considered learners and tuning choices relative to their statistical uncertainty. Complete re-estimation after the outcome translation leaves the honest-forest, OLS/logit, and `ranger` estimates numerically unchanged. The tuned XGBoost estimates move slightly, with a largest difference of 0.0029. The reconstructed XGBoost weights do not satisfy the exact representation check, so their weight diagnostics are interpreted with this qualification.

The outcome-weight analysis explains the kappa result more directly. Under cubic age adjustment, translation sensitivity occurs when the treated and control weight masses are not both equal to one. At the same time, the effective sample sizes and maximum weights remain similar across the kappa estimators. Normalisation and weight concentration therefore describe different features of the estimators. The balance results add a further distinction. All selected estimators reduce the age differences, but exact balance depends on the propensity-score specification and is not implied by translation invariance.

The next application studies Card's college-proximity design using two binary schooling margins and two covariate specifications.

8 Empirical Application: Returns to Schooling

8.1 Data and Design

This application revisits **?**, who uses geographic proximity to a four-year college as an instrument for schooling. Direct wage comparisons do not identify a causal effect of schooling when education choices are related to unobserved determinants of wages. The instrument exploits variation in the cost of college attendance. Growing up near a four-year college can reduce relocation and commuting costs and may therefore increase post-secondary schooling (**?**).

8.1.1 Sample, Variables, and Treatment Definitions

Following **?**, the analysis uses the National Longitudinal Survey of Young Men. The supplied analysis file contains N = 3,010 men. No observation is removed by the non-missing wage and education restriction, and the variables used below are complete in this file. The outcome Y_i is log hourly wage. The instrument Z_i indicates whether the individual grew up near a four-year college in 1966.

The empirical design uses two binary schooling margins. The first treatment is

$$D_{1i} = 1\{\texttt{educ}_i > 12\},$$

which records schooling beyond high school. The second treatment is

$$D_{2i} = 1\{\texttt{educ}_i \geq 16\},$$

which records at least sixteen completed years of schooling. The two treatments therefore define distinct conditional LATE parameters rather than a return to one additional year of schooling. Appendix Table **??** lists the instrument, treatments, and outcome.

8.1.2 Covariate Specifications

The identifying restriction is conditional. In the notation of Section **??**, it requires instrument independence given X_i, together with exclusion and monotonicity. These assumptions are not established by the descriptive evidence below. Residential sorting and local labour-market differences remain possible threats to the interpretation of college proximity as an instrument.

Following **?**, I use two conditioning sets. The Card specification includes experience, experience squared, indicators for Black, South and SMSA residence in 1976, SMSA residence in 1966, and the regional indicators `reg661`, ..., `reg668`. The parsimonious Kitagawa specification includes indicators for Black, South and SMSA residence in 1976, SMSA residence in 1966, and South residence in 1966 (?). The content of X_i is part of the identifying restriction, so both specifications are retained throughout the application. Appendix Table **??** records the exact variables.

8.2 Descriptives

The descriptive analysis focuses on two features that enter the later weighting estimators directly. The first is the conditional association between the instrument and each treatment. The second is support in the estimated instrument propensity score.

For schooling beyond high school, the treatment rate is 0.544 among individuals with $Z_i = 1$ and 0.422 among individuals with $Z_i = 0$. The unadjusted difference is 0.122. For the sixteen-year margin, the corresponding rates are 0.293 and 0.225, which gives a smaller unadjusted difference of 0.069. These are treatment-rate contrasts. They are not labelled as complier or principal-strata shares because the empirical design conditions on X_i.

Table **??** reports the covariate-adjusted first stages. For D_{1i}, the estimated coefficient falls to 0.064 under the Card specification and 0.065 under the Kitagawa specification. The associated squared HC1-robust t-statistics are 12.5 and 9.0. The evidence is therefore somewhat stronger for the Card conditioning set. For D_{2i}, the adjusted first stages are only 0.030 and 0.038, with squared robust t-statistics of 3.5 and 3.9. This is the clearest design warning in the Card application. The college-proximity instrument contains substantially less conditional variation for the sixteen-year schooling margin. The reported statistics are descriptive strength diagnostics and are not a weak-instrument-robust test.

Table 3: Unadjusted and covariate-adjusted first-stage diagnostics

Treatment	Specification	Unadjusted gap	$\hat{\delta}$	HC1 SE	Robust t^2
`somecol`	Card	0.1219	0.0636	0.0180	12.5
`somecol`	Kitagawa	0.1219	0.0653	0.0218	9.0
`educ16`	Card	0.0686	0.0302	0.0162	3.5
`educ16`	Kitagawa	0.0686	0.0379	0.0191	3.9

Notes: The unadjusted gap is $\Pr(D = 1 \mid Z = 1) - \Pr(D = 1 \mid Z = 0)$. The adjusted coefficient $\hat{\delta}$ is obtained from a linear probability model using the covariates in the stated specification. HC1 SE is the heteroskedasticity-robust standard error. Robust t^2 is the squared HC1-robust t-statistic for Z_i.

The fitted instrument propensity score is $\hat{p}(X_i) = \widehat{\Pr}(Z_i = 1 \mid X_i)$. It is directly relevant for the kappa estimators because their weights contain inverse propensity terms. The relevant tail depends on the observed instrument value. Large values of $\hat{p}(X_i)$ are most consequential among observations with $Z_i = 0$, where $1/(1 - \hat{p}(X_i))$ can become large.

Under the Card specification, the fitted scores range from 0.172 to 0.952. Among the 957 observations with $Z_i = 0$, the maximum is 0.936, and 40 observations have a fitted score above 0.90. Under the Kitagawa specification, the maximum among $Z_i = 0$ is 0.907, and only 9 observations exceed 0.90. No observation with $Z_i = 1$ has a fitted score below 0.10 under either specification. The support issue is therefore concentrated in the high-propensity tail of the $Z_i = 0$ group, particularly under the Card specification. This finding motivates the later weight concentration diagnostics, but it does not determine the final outcome weights on its own. Appendix Table **??** and Appendix Figure **??** provide the full propensity summary and distribution.

r IV score. Moroever, the Wald-AIPW, which combines a binary-instrument Wald contrast with augmented inverse-propensity components Table **??** shows a clear separation between PLR-IV and Wald-AIPW. PLR-IV is close to the conventional 2SLS benchmark. For the `somecol` margin, PLR-IV gives 0.681 under the Card specification and 0.593 under the Kitagawa specification, almost identical to the corresponding 2SLS estimates. The same pattern appears for `educ16`, where PLR-IV remains close to the larger 2SLS estimates. This is consistent with the interpretation of PLR-IV as a flexible version of a partially linear IV, rather than as a direct analogue of the normalized kappa LATE estimators. On the other hand, the Wald AIPW behaves different. The grf and ranger implementations are substantially smaller than PLR-IV and closer to the normalized kappa benchmark. For `somecol`, both forest-based Wald-AIPW estimates lie around 0.26–0.29, compared with normalized kappa estimates of 0.331 and 0.356. For `educ16`, the forest-based Wald-AIPW estimates are around 0.49–0.54, again below PLR-IV and 2SLS but close to the normalized kappa estimates. These estimates are very close to the $\hat{\tau}_u^{ml}$ but more conservative then the PLR-IV. The parametric linear/logit Wald-AIPW estimator provides an intermediate benchmark. For `somecol`, it is close to the normalized kappa and forest-based Wald-AIPW estimates. For `educ16`, however, it is larger under the Card specification. This could be due to the fact that the parametric learner imposes stronger functional-form restrictions on the nuisance functions. The XGBoost row is qualitatively different. Its estimates are not only larger or smaller than the other DML estimates, but they are outside the empirical range of all other estimators and even change sign. This again showcases evidence that the Wald-AIPW score is highly sensitive to the nuisance learner in this application. The score depends on several estimated nuisance components, including instrument propensities and conditional outcome and treatment regressions. In a weak-first-stage design, small errors in these components can strongly affect the final Wald ratio. Sequential boosting can amplify this problem if the fitted nuisance functions are poorly calibrated in regions with limited overlap.

8.2.1 Nested XGBoost tuning as a Robustness check

The instability of the default XGBoost specification motivates a nested tuning robustness check. The purpose of this exercise is not to search for a treatment-effect estimate that

Table 4: DML-IV estimates in the Card application

	`somecol`		`educ16`	
Estimator	Card	Kitagawa	Card	Kitagawa
2SLS	0.661	0.575	1.392	0.991
$\hat{\tau}_u^{ml}$ (kappa)	0.331	0.356	0.619	0.628
PLR-IV (grf / OutcomeWeights)	0.681	0.593	1.140	1.030
Wald-AIPW (grf / OutcomeWeights)	0.279	0.292	0.530	0.535
Wald-AIPW (linear/logit)	0.399	0.311	0.884	0.534
Wald-AIPW (ranger)	0.262	0.294	0.493	0.514
Wald-AIPW (XGBoost)	-13.416	-0.113	-3.198	-3.656

looks plausible. Instead, the XGBoost hyperparameters are selected only by predictive nuisance performance inside the DML procedure. The final treatment-effect estimate is not used as a tuning criterion, as for the continuous outcome nuisance, tuning minimizes root mean squared error and for the binary treatment and instrument nuisances, tuning minimizes log-loss. Table **??** reports the resulting untuned and tuned XGBoost Wald-AIPW estimates. The untuned results reproduce the extreme XGBoost rows in Table **??**. After tuning, the estimates move back into the range of the other Wald-AIPW and normalized kappa estimates. For the Card specification, the somecol estimate changes from (-13.416) to (0.352), while the educ16 estimate changes from (-3.198) to (0.627). The same pattern appears under the Kitagawa specification, where the tuned estimates are (0.258) for somecol and (0.475) for educ16. The tuning results suggest that the extreme default XGBoost estimates are largely driven by nuisance calibration rather than by a stable causal signal. In the Card specification, the default XGBoost learner produces many boundary propensity predictions, while the tuned specification removes them under the chosen threshold. This supports the interpretation that the original XGBoost row is a diagnostic warning about learner instability. At the same time, the tuned XGBoost estimates should still be interpreted as a sensitivity check rather than as the preferred estimator. Overall, flexible nuisance estimation does not remove the empirical difficulty of this design. PLR-IV remains close to 2SLS, while Wald-AIPW with grf, ranger, and the parametric learner is closer to the normalized kappa estimates. The default XGBoost specification is the exception and is best interpreted as a learner-instability diagnostic. The nested tuning exercise shows that this instability is strongly reduced when XGBoost is selected by nuisance-prediction loss.

8.3 Estimates across Treatment Margins and Conditioning Sets

Table **??** reports a selected set of estimates for the four combinations of treatment and covariate specification. The table keeps the two score families separate. PLR-IV is based on the partially linear IV score, while Wald-AIPW uses the binary-instrument Wald score

Table 5: Nested XGBoost nuisance tuning: Wald-AIPW estimates

Cell	XGB untuned	XGB tuned	Propensity boundary share
Card–somecol	(-13.416) ((14.411))	(0.352) ((0.228))	(0.2595 $\rightarrow$ 0.0000)
Card–educ16	(-3.198) ((4.054))	(0.627) ((0.420))	(0.2595 $\rightarrow$ 0.0000)
Kitagawa–somecol	(-0.113) ((0.627))	(0.258) ((0.269))	(0.0017 $\rightarrow$ 0.0000)
Kitagawa–educ16	(-3.656) ((100.914))	(0.475) ((0.492))	(0.0017 $\rightarrow$ 0.0000)

with augmented inverse-propensity terms. The honest-forest and XGBoost results use the tuned implementations described in the common empirical protocol. Hyperparameters are selected using only out-of-sample nuisance-prediction loss. The treatment-effect estimates do not enter the tuning criterion.

Table 6: Selected estimates in the Card application

	Schooling beyond high school		At least 16 years	
Estimator	Card	Kitagawa	Card	Kitagawa
2SLS	0.661	0.575	1.392	0.991
	(0.295)	(0.308)	(0.801)	(0.611)
$\hat{\tau}_u^{cb}$	0.376	0.331	0.853	0.588
	(0.223)	(0.236)	(0.549)	(0.433)
$\hat{\tau}_u^{ml}$	0.331	0.356	0.619	0.628
	(0.202)	(0.244)	(0.387)	(0.448)
$\hat{\tau}_t^{ml}$	0.171	0.769	0.319	1.367
	(0.367)	(0.308)	(0.687)	(0.648)
PLR-IV, tuned forest	0.681	0.593	1.140	1.030
	(0.289)	(0.340)	(0.535)	(0.684)
Wald-AIPW, tuned forest	0.279	0.292	0.530	0.535
	(0.214)	(0.249)	(0.398)	(0.462)
PLR-IV, tuned XGBoost	0.577	0.612	0.956	1.010
	(0.258)	(0.350)	(0.459)	(0.655)
Wald-AIPW, tuned XGBoost	0.382	0.238	0.730	0.477
	(0.214)	(0.258)	(0.429)	(0.514)

Notes: The outcome is log hourly wage measured in dollars. Standard errors are shown in parentheses. The kappa superscript identifies the propensity-score implementation and the subscript identifies the normalisation. The XGBoost rows use fold-specific nested tuning based on nuisance-prediction loss. Appendix Tables **??** and **??** report the remaining estimators and learner specifications.

All selected estimates are positive, but the estimates for the sixteen-year margin are less precise. This is consistent with the weaker conditional first stage documented in Table **??**. A larger coefficient for this margin should therefore not be read as more reliable evidence of a larger effect. Two further patterns are visible across all four design cells. First, the PLR-IV estimates are numerically closer to 2SLS, whereas the Wald-AIPW estimates are closer to the normalised kappa estimates. This pattern remains when honest forests are replaced by tuned XGBoost. It is a comparison of finite-sample estimates, not evidence that the estimators have the same target or that one set provides the correct benchmark. Second, the normalised kappa and tuned DML estimates change moderately when the conditioning set changes. The representative unnormalised estimator $\hat{\tau}_t^{ml}$ is more sensitive. For schooling beyond high school it changes from 0.171 under the Card specification to 0.769 under the Kitagawa specification. For the sixteen-year margin it changes from 0.319 to 1.367. The next subsection examines whether this difference is linked to the response of the estimators to a pure outcome translation.

8.4 Translation Invariance under Complete Re-estimation

Following the protocol in Section **??**, every estimator is fitted again after adding log(100). The normalised kappa estimators $\hat{\tau}_u^{cb}$, $\hat{\tau}_u^{ml}$, and $\hat{\tau}_{a,10}^{ml}$ reproduce exactly in all four cells. The three unnormalised kappa estimators fail in every cell. For $\hat{\tau}_t^{ml}$, the change equals -0.492 and -0.921 under the Card specification, and 1.284 and 2.283 under the Kitagawa specification. The different signs are not random numerical noise. They correspond to whether the untreated component receives more or less than unit mass.

The DML results require a more qualified interpretation. Linear PLR-IV and linear/logit Wald-AIPW reproduce exactly. The tuned honest forests differ by at most 0.00324. They pass the numerical tolerance under the Kitagawa specification but not under the Card specification. Tuned XGBoost is also not exactly invariant. Its largest difference is 0.02183 for PLR-IV under the Card sixteen-year specification and 0.00844 for Wald-AIPW under the Kitagawa sixteen-year specification. These differences are small relative to the corresponding standard errors, but the exact finite-sample equality is not obtained.

The rerun test is a finite-sample property of an estimator and its implementation. It does not test instrument validity and it does not rank estimators by causal accuracy. Its value here is narrower. It separates the exact normalisation built into the normalised kappa formulas from the approximate behaviour of flexible DML implementations after the complete estimation procedure is repeated.

8.5 Outcome Weights and Covariate Balance

The normalised kappa estimators assign unit mass to the treated component and minus unit mass to the untreated component, so their total weight is zero. By contrast, $\hat{\tau}_t^{ml}$ has treated mass equal to one but sign-adjusted untreated masses of 1.11, 1.20, 0.72, and 0.50 across the four cells. These deviations match the direction and size of its translation changes.

Weight normalisation is distinct from weight concentration. The selected normalised kappa and tuned DML estimators retain modified effective sample sizes above 1,300, and the tuned forest weights have treated and sign-adjusted untreated masses close to one. The sixteen-year margin tends to have larger maximum absolute weights, especially for

Wald-AIPW under the Card specification, but the modified effective sample size is scale invariant and does not necessarily fall when the treatment definition changes. Appendix Table **??** reports the selected mass and concentration measures.

Covariate balance is also a separate diagnostic. The Love plots compare the treatment groups entering each weighted contrast. They do not compare the instrument groups and cannot establish the conditional IV assumptions. The CBPS-based normalised kappa estimator achieves near-exact balance in the included covariates by construction. Other propensity-score and learner choices leave different amounts of residual imbalance. The figures are reported in Appendix Section **??**. They are treated as supplementary implementation diagnostics rather than as a test of instrument validity.

Overall, the Card design yields three findings. The sixteen-year treatment margin has the weaker conditional first stage. The gap between PLR-IV and Wald-AIPW persists across the two conditioning sets and the tuned learner families. Most importantly for the research question, exact translation invariance is obtained by the normalised kappa estimators but not by their unnormalised counterparts. Flexible DML implementations are much closer to invariance than the unnormalised kappa estimators, although complete re-estimation does not always produce exact equality.

9 Empirical Child Application

A third central question in empirical labor economics is how child birth or the birth of an additional child affect mothers' labor-market outcomes. Families with more children often differ from families with fewer children in many observed and unobserved dimensions. Those can include preferences for work and family size, household resources and local labor-market conditions. A simple comparison between women with two children and women with more than two children would therefore not isolate the causal effect of childbearing. It would combine the effect of an additional child with selection into larger families. The third design based on the application of **?** addresses this selection problem by using sibling sex composition as an instrument for fertility. This application complements the two previous empirical settings. The Vietnam draft lottery provides a low-dimensional, near-random-assignment benchmark, while the Card application studies an observational instrument with a richer covariate structure. The application based on **?** instead combines a very large parent sample with binary labor-supply and log-income outcomes. It is therefore especially useful for the translation-invariance analysis, because the outcome coding of both labor-force participation and income is directly relevant for the behavior of normalized and unnormalized estimators. The next subsection introduces the data, the treatment and instrument definitions, and the two outcome variables used in the analysis.

9.1 Data and Design

The analysis follows **?** in revisiting the **?** application based on the 1980 U.S. Census. The sample consists of women with at least two children. The empirical question is whether having an additional child affects mothers' labor-market outcomes. The treatment is therefore an indicator for whether a woman has more than two children,

$$D_i = \mathbf{1}\Big\{N_i^{\text{children}} > 2\Big\}\,.$$

The identifying variation comes from the sex composition of the first two children. Following **?**, the instrument is an indicator for whether the first two children are of the same sex,

$$Z_i = \mathbf{1}\{\text{boy}_{1i} = \text{boy}_{2i}\}\,,$$

where boy_{1i} and boy_{2i} indicate whether the first and second child are boys. Equivalently, $Z_i = 1$ if the first two children are either two boys or two girls, and $Z_i = 0$ if the first two children are of mixed sex. The economic mechanism is that parents with two children of the same sex are more likely to have a third child, because some parents have a preference for a mixed-sex sibling composition. The instrument therefore shifts the probability of further childbearing but does not deterministically assign treatment.

I consider two outcome variables. The first is labor-force participation, defined as an indicator for whether the woman worked for pay in the preceding year,

$$Y_i^{\text{LFP}} = \mathbf{1}\{\text{worked}_i = 1\} .$$

The second is log annual income,

$$Y_i^{\text{INC}} = \log(\text{income}_i) ,$$

which is only defined for women with strictly positive reported income. These two outcomes are useful for different parts of the analysis. For the analysis, I use the following covariates, following the specification in **?**, common to both outcome samples and all estimators. Table **??** lists them.

Table 7: Covariates used in the Angrist–Evans application

Covariate	Definition
age_i	Mother's age at census observation
agefirst_i	Mother's age at first birth
boy_{1i}	Indicator that the first child is a boy
boy_{2i}	Indicator that the second child is a boy
Black_i	Race indicator: Black
Hispanic_i	Ethnicity indicator: Hispanic
OtherRace_i	Race indicator: other race

9.1.1 Subsampling for the Angrist–Evans Application

Following **?**, I use two corresponding parent samples from **?** 1980 Census extract. I use two corresponding parent samples. The first is the labor-force participation sample, defined for all 394,840 women with at least two children. The second is the log-income sample, restricted to the 220,502 women with strictly positive reported income. The full parent samples are too large for the computationally intensive diagnostics used below, especially for the estimation of the Smoother matrices. I therefore construct smaller diagnostic subsamples. Their purpose is to preserve the central empirical structure of the corresponding parent samples, in particular the instrument and treatment shares, the first-stage structure, and the relevant outcome distributions. To this end, I use proportional stratified subsampling without replacement. The procedure is applied separately to the labor-force participation and log-income parent samples, using a fixed seed. Each resulting diagnostic subsample contains 3,000 observations. The exact stratification variables and allocation rule are described in Appendix **??**.

9.2 Descriptive Diagnostics

Before estimating the weighted LATE estimators, I first inspect the basic empirical structure of the two created subsamples. Figure **??** reports the marginal shares of the instrument and the treatment separately for the labor-force participation and log-income samples.

Both subsamples, display instrument shares of 50.5 percent in the labor-force sample and 50.3 percent in the log-income sample. This near equal split is consistent with the random nature of child sex composition. The treatment shares differ more clearly across samples. In the labor-force participation sample, 40.2 percent of women have more than two children. In the positive-income sample, the corresponding share is only 34.8 percent.

pics/ae_instrument_treatment_shares.png

Figure 1: Instrument and treatment shares in the Angrist–Evans diagnostic subsamples

These marginal shares describe the composition of the two subsamples. They do not yet show whether the instrument shifts fertility behavior. The relevant first-stage question is whether women with same-sex first two children are more likely to have more than two children than women with mixed-sex first two children. Hence, Table **??** reports the raw first-stage, reduced-form, and Wald contrasts. In both subsamples, women whose first two children have the same sex are more likely to have more than two children. In the labor-force participation sample, the treatment rate increases from 37.1 percent among mixed-sex families to 43.2 percent among same-sex families. This implies a raw first stage of 6.1 percentage points. Similar in the log-income sample, a raw first stage of 5.4 percentage points is visible. The positive first stage supports the relevance of the same-sex instrument. The magnitude is economically meaningful but not large. The reduced forms are negative in both samples. In the labor-force participation sample, same-sex families have a labor-force participation rate that is 0.7 percentage points lower than mixed-sex families. In the log-income sample, the reduced form is also negative, with a difference of about 0.012 log points. Since the first stage is positive in both samples, these negative reduced forms imply negative raw Wald ratios. This underlines as well the exclusion restriction.

Table 8: Raw first-stage, reduced-form, and Wald contrasts

	Labor-force sample	Log-income sample
$\Pr(D_i = 1 \mid Z_i = 0)$	0.371	0.321
$\Pr(D_i = 1 \mid Z_i = 1)$	0.432	0.374
First stage	0.061	0.053
Reduced form	-0.007	-0.012
Raw Wald ratio	-0.121	-0.230

The raw IV diagnostics show that the same-sex instrument shifts fertility behavior in both diagnostic subsamples. Since the kappa-based estimators use the instrument propensity score, I next examine the distribution of the instrument propensity score. Compared to the prior frameworks in this design, severe propensity-score overlap problems are not expected. The instrument is based on sibling sex composition and should therefore be close to randomly assigned. Figure **??** reflects this. In both subsamples, the fitted propensity scores are tightly concentrated around one half. The labor-force sample ranges from approximately 0.45 to 0.56, while the log-income sample ranges from approximately 0.46 to 0.58. There are no observations close to zero or one. This differs pretty much to the prior college proximity application In the prior application college proximity is an observational instrument and the estimated propensity score is more dispersed. In the present design, the instrument propensity score behaves much more like in a randomized setting.

pics/ae_pscore_overlap.png

Figure 2: Estimated instrument propensity-score distributions in the Angrist–Evans diagnostic subsamples

As both sets are subsamples of a bigger dataset a short examination of the covariate balance by instrument status is conducted. Since the same-sex instrument is based on the sex composition of the first two children, large systematic differences in predetermined maternal characteristics would be unexpected. The balance diagnostics reported in Appendix **??** are consistent with this interpretation. Standardized mean differences are small in both diagnostic subsamples, suggesting that women with same-sex and mixed-sex first two children are similar along the observed covariates used in the propensity-score model. This supports the plausibility of the conditional independence assumption, although it cannot test the exclusion restriction directly. Finally, one technical feature of this design is important for the later nuisance estimation. The instrument is constructed from the sex indicators of the first and second child. Therefore, these two covariates are almost uncorrelated in the pooled sample, but become mechanically related once one conditions on the instrument. Among same-sex families, the two indicators are identical. Among mixed-sex families, they are exact opposites. Hence, within the $Z = 1$ arm their correlation is one, while within the $Z = 0$ arm their correlation is minus one.

Table 9: Structural relation between first- and second-child sex indicators

Sample	Panel	$\mathrm{cor}(\mathrm{sexk}, \mathrm{sex2nd})$
Labor-force sample	Pooled	0.010
Labor-force sample	$Z = 0$	-1.000
Labor-force sample	$Z = 1$	1.000
Log-income sample	Pooled	0.005
Log-income sample	$Z = 0$	-1.000
Log-income sample	$Z = 1$	1.000

This relationship is not an empirical imbalance. It follows directly from the definition of the instrument. However, it matters for later parametric nuisance models. In pooled propensity-score models, both child-sex indicators can be included as separate main effects. In arm-specific outcome or treatment models, however, including both indicators together with an intercept would create perfect collinearity. For this reason, the arm-specific parametric nuisance specifications used below omit one of the two child-sex indicators.

9.3 Reproduction of the SUW Recoding Table

? report the Angrist–Evans application in a single table with seven outcome codings. These consist of three codings of labor-force participation and four unit codings of log income. Since the present thesis uses diagnostic subsamples of 3,000 observations for each outcome family, I reproduce the same structure in two separate tables. Table **??** reports the labor-force recodings and Table **??** reports the income unit recodings. Because this thesis uses smaller diagnostic subsamples, the purpose is not to match the full-sample point estimates exactly.

The normalized estimators in Panel B behave exactly as translation invariance predicts. Within each row, the estimates are identical up to rounding across all pure outcome translations. For labor-force participation, the first two columns compare $Y_i = 1\{\mathrm{worked}_i = 1\}$ with $Y_i + 1$. For income, changing the unit of measurement before taking logarithms only adds a constant to the logged outcome.

The third labor-force column requires a separate comment. The coding $1\{\text{did not work}\} = 1 - 1\{\text{worked}\}$ is not a pure translation. It is a reflection plus a shift, $Y_i' = -Y_i + 1$. **?** explicitly state in their table note that, for this coding, they report the additive inverse of the raw estimate. Table **??** follows this display convention: only the reported point estimate in the third labor-force column is multiplied by -1, while the standard error is unchanged. This convention is used only to match the published SUW layout. It is not applied to the income table, to the DML estimates below, or to the later translation-invariance rerun analysis.

The unnormalized estimators in Panel C are visibly sensitive to the outcome coding. In the labor table, for example, $\hat{\tau}_a^{ml}$ equals -0.091 under the original coding, -0.083 under the shifted coding, and -0.099 under the display-adjusted reflected coding. The implied realized outcome-weight sum of about 0.008 comes from comparing the first two columns, which are the genuine translation Y_i and $Y_i + 1$. The third column is consistent with the same value only after undoing the SUW display convention: the displayed value -0.099 corresponds to a raw reflected estimate of $+0.099$, and the reflection formula then gives the same implied weight sum.

The diagnostic subsamples reproduce the qualitative pattern of **?**: normalized estimators are invariant to recoding, while unnormalized estimators are not. The magnitude of the sensitivity is smaller here than in the published full-sample table, but this is not problematic because the tables are based on different samples. The main point estimates are also close to the published full-sample estimates relative to the uncertainty induced by the smaller diagnostic samples. For example, the labor 2SLS estimate is -0.096 in the diagnostic subsample, compared with -0.117 in **?**. For log income, the corresponding estimates are -0.170 and -0.135. These differences are small relative to the standard errors in the diagnostic subsamples. As a rough sampling-scale benchmark, scaling the published standard errors by the square root of the sample-size ratio gives $0.025\sqrt{394{,}840/3{,}000} \approx 0.287$ for labor and $0.092\sqrt{220{,}502/3{,}000} \approx 0.789$ for income, close to the observed diagnostic-subsample standard errors of 0.275 and 0.752. The point estimates and standard errors should therefore be interpreted as subsample diagnostics. They are useful for showing that the same finite-sample mechanism is present, not for replacing the full-sample estimates reported by **?**

Table 10: AE98 reproduction — labor-force participation

Estimator	1 if worked, 0 else		2 if worked, 1 else		1 if did not work, 0 else	
Panel A: 2SLS (HC1-robust)						
2SLS	−0.096	(0.275)	−0.096	(0.275)	−0.096	(0.275)
Panel B: Normalized kappa estimators						
$\hat{\tau}_u^{cb}$	−0.095	(0.274)	−0.095	(0.274)	−0.095	(0.274)
$\hat{\tau}_u^{ml}$	−0.095	(0.274)	−0.095	(0.274)	−0.095	(0.274)
$\hat{\tau}_{a,10}^{ml}$	−0.095	(0.275)	−0.095	(0.275)	−0.095	(0.275)
Panel C: Unnormalized kappa estimators						
$\hat{\tau}_a^{ml}$	−0.091	(0.275)	−0.083	(0.277)	−0.099	(0.275)
$\hat{\tau}_{a,1}^{ml}$	−0.091	(0.274)	−0.082	(0.276)	−0.099	(0.273)
$\hat{\tau}_{a,0}^{ml}$	−0.091	(0.276)	−0.083	(0.278)	−0.099	(0.276)

Notes: Diagnostic subsample, N = 3,000; **?** report N = 394,840 on the full sample. Each cell reports the point estimate with the standard error below. Covariates as in Table **??**. Following **?**, the third column reports the additive inverse of the raw estimate; see the discussion below.

Table 11: AE98 reproduction — log family income

Estimator	Cents	Dollars	\$1000s	\$100,000s
Panel A: 2SLS (HC1-robust)				
2SLS	−0.170(0.752)	−0.170(0.752)	−0.170(0.752)	−0.170(0.752)
Panel B: Normalized kappa estimators				
$\hat{\tau}_u^{cb}$	−0.168(0.751)	−0.168(0.751)	−0.168(0.751)	−0.168(0.751)
$\hat{\tau}_u^{ml}$	−0.168(0.751)	−0.168(0.751)	−0.168(0.751)	−0.168(0.751)
$\hat{\tau}_{a,10}^{ml}$	−0.167(0.754)	−0.167(0.754)	−0.167(0.754)	−0.167(0.754)
Panel C: Unnormalized kappa estimators				
$\hat{\tau}_a^{ml}$	0.011 (0.848)	−0.050(0.796)	−0.142(0.756)	−0.203(0.756)
$\hat{\tau}_{a,1}^{ml}$	0.011 (0.843)	−0.050(0.791)	−0.141(0.751)	−0.201(0.751)
$\hat{\tau}_{a,0}^{ml}$	0.011 (0.854)	−0.050(0.802)	−0.143(0.760)	−0.204(0.761)

Notes: Diagnostic subsample, N = 3,000; **?** report N = 220,502 on the full sample. Each cell reports the point estimate with the standard error below. Covariates as in Table **??**.

10 Discussion

10.1 Differences in the design of the applications

The Card application is useful for this thesis because it changes the empirical design relative to the Vietnam draft lottery. In the Vietnam application, the instrument was close to randomly assigned conditional on age, and the covariate adjustment problem was deliberately low-dimensional. In the Card application, by contrast, college proximity is observational. The credibility of the instrument depends on conditioning on individual, family, regional, and labour-market characteristics. The instrument propensity score is therefore no longer nearly flat, and the choice of covariate specification becomes part of the empirical problem.

10.2 Cross-Application Summary

10.3 The Outcome-Weights Lens: What It Adds

10.4 DML Learner Comparison and Implications

10.5 Practical Guidance

10.6 Limitations

11 Conclusion

12 Appendix

A Normalization of the Weighting Estimators

This appendix shows explicitly how the weights entering $\widehat{\tau}_u$ and $\widehat{\tau}_{a,10}$ are normalized.

A.1 Normalization of $\widehat{\tau}_u$

Define the normalized inverse-probability weights for observations with $Z_i = 1$ and $Z_i = 0$, respectively, as

$$h_{1i} = \frac{Z_i/p(X_i)}{\sum_{j=1}^{N} Z_j/p(X_j)}, \qquad h_{0i} = \frac{(1-Z_i)/(1-p(X_i))}{\sum_{j=1}^{N}(1-Z_j)/(1-p(X_j))}. \tag{85}$$

By construction,

$$\sum_{i=1}^{N} h_{1i} = \frac{\sum_{i=1}^{N} Z_i/p(X_i)}{\sum_{j=1}^{N} Z_j/p(X_j)} = 1, \tag{86}$$

and

$$\sum_{i=1}^{N} h_{0i} = \frac{\sum_{i=1}^{N}(1-Z_i)/(1-p(X_i))}{\sum_{j=1}^{N}(1-Z_j)/(1-p(X_j))} = 1. \tag{87}$$

Using these weights, define the normalized weighted means

$$\widehat{\mu}_{Y,1} = \sum_{i=1}^{N} h_{1i} Y_i, \qquad \widehat{\mu}_{Y,0} = \sum_{i=1}^{N} h_{0i} Y_i, \tag{88}$$

$$\widehat{\mu}_{D,1} = \sum_{i=1}^{N} h_{1i} D_i, \qquad \widehat{\mu}_{D,0} = \sum_{i=1}^{N} h_{0i} D_i. \tag{89}$$

The estimator in Equation (**??**) can therefore be written compactly as

$$\widehat{\tau}_u = \frac{\widehat{\mu}_{Y,1} - \widehat{\mu}_{Y,0}}{\widehat{\mu}_{D,1} - \widehat{\mu}_{D,0}}. \tag{90}$$

Thus, $\widehat{\tau}_u$ is a weighted Wald estimator in which both the reduced-form contrast and the first-stage contrast are constructed from separately normalized inverse-probability-weighted means. The reduced-form numerator may also be written as a single weighted sum:

$$\widehat{\mu}_{Y,1} - \widehat{\mu}_{Y,0} = \sum_{i=1}^{N} (h_{1i} - h_{0i}) Y_i. \tag{91}$$

The corresponding signed outcome weights sum to zero:

$$\sum_{i=1}^{N} (h_{1i} - h_{0i}) = \sum_{i=1}^{N} h_{1i} - \sum_{i=1}^{N} h_{0i} = 1 - 1 = 0. \tag{92}$$

Hence, although the weights within each component sum to one, the signed weights entering their difference sum to zero.

A.2 Normalization of $\widehat{\tau}_{a,10}$

The estimator $\widehat{\tau}_{a,10}$ is given by

$$\widehat{\tau}_{a,10} = \frac{\sum_{i=1}^{N} \kappa_{1i} Y_i}{\sum_{i=1}^{N} \kappa_{1i}} - \frac{\sum_{i=1}^{N} \kappa_{0i} Y_i}{\sum_{i=1}^{N} \kappa_{0i}}. \tag{93}$$

Define the normalized component weights

$$a_i = \frac{\kappa_{1i}}{\sum_{j=1}^{N} \kappa_{1j}}, \qquad b_i = \frac{\kappa_{0i}}{\sum_{j=1}^{N} \kappa_{0j}}. \tag{94}$$

Provided that the respective denominators are nonzero, these weights satisfy

$$\sum_{i=1}^{N} a_i = \frac{\sum_{i=1}^{N} \kappa_{1i}}{\sum_{j=1}^{N} \kappa_{1j}} = 1, \qquad \sum_{i=1}^{N} b_i = \frac{\sum_{i=1}^{N} \kappa_{0i}}{\sum_{j=1}^{N} \kappa_{0j}} = 1. \tag{95}$$

Accordingly,

$$\widehat{\tau}_{a,10} = \sum_{i=1}^{N} a_i Y_i - \sum_{i=1}^{N} b_i Y_i. \tag{96}$$

The estimator can equivalently be expressed as a single signed weighted sum:

$$\widehat{\tau}_{a,10} = \sum_{i=1}^{N} (a_i - b_i) Y_i. \tag{97}$$

Defining

$$\omega_i = a_i - b_i, \tag{98}$$

the signed outcome weights satisfy

$$\sum_{i=1}^{N} \omega_i = \sum_{i=1}^{N} a_i - \sum_{i=1}^{N} b_i = 1 - 1 = 0. \tag{99}$$

It is therefore important to distinguish between two statements. First, the weights used in each component of $\widehat{\tau}_{a,10}$ sum separately to one. Second, once the two components are combined into a treatment-effect contrast, the resulting signed outcome weights sum to zero.

B PLR-IV as an aggregation of conditional Wald estimands

The difference between the PLR-IV estimand and the unconditional LATE can be made explicit when the instrument is binary. Let

$$p(X) = P(Z = 1 \mid X)$$

and define the conditional reduced form and first stage as

$$\Delta_Y(X) = E[Y \mid Z = 1, X] - E[Y \mid Z = 0, X], \tag{100}$$

$$\Delta_D(X) = E[D \mid Z = 1, X] - E[D \mid Z = 0, X]. \tag{101}$$

Since

$$E[(Z - p(X))Y \mid X] = p(X)\{1 - p(X)\}\Delta_Y(X)$$

and

$$E[(Z - p(X))D \mid X] = p(X)\{1 - p(X)\}\Delta_D(X),$$

the residualized IV estimand can be written as

$$\theta_{\mathrm{PLR-IV}} = \frac{E[p(X)\{1 - p(X)\}\Delta_Y(X)]}{E[p(X)\{1 - p(X)\}\Delta_D(X)]}. \tag{102}$$

Under the conditional IV assumptions and monotonicity, $\Delta_D(X)$ equals the conditional complier share. Whenever $\Delta_D(X) > 0$, the conditional Wald ratio

$$\tau_C(X) = \frac{\Delta_Y(X)}{\Delta_D(X)}$$

identifies the conditional LATE. Equation (**??**) can therefore be rewritten as

$$\theta_{\mathrm{PLR-IV}} = \frac{E[p(X)\{1 - p(X)\}\Delta_D(X)\tau_C(X)]}{E[p(X)\{1 - p(X)\}\Delta_D(X)]}. \tag{103}$$

By contrast, the unconditional LATE is

$$\tau_{\mathrm{LATE}} = \frac{E[\Delta_D(X)\tau_C(X)]}{E[\Delta_D(X)]}. \tag{104}$$

Thus, PLR-IV attaches additional weight to covariate regions in which the instrument propensity is close to one half. The two aggregations generally differ when $p(X)$ and the conditional complier effect vary across covariate values. They coincide, for example, when the conditional LATE is constant or when $p(X)$ is constant across X.

C Outcome-Weight Classification

Table ?? reproduces the classification introduced by ?, Table 4 using the notation of this thesis.

Table 12: Outcome-weight classification of Knaus (2024)

Weight class	$C = \sum_i \omega_i$	$C_1 = \sum_i \omega_i D_i$	$C_0 = \sum_i \omega_i (1 - D_i)$
Fully unnormalized	$\neq 0$	$\neq 1$	$\neq -1$
Untreated unnormalized	$\neq 0$	$= 1$	$\neq -1$
Treated unnormalized	$\neq 0$	$\neq 1$	$= -1$
Scale-normalized	$= 0$	$= c \neq 1$	$= -c \neq -1$
Fully normalized	$= 0$	$= 1$	$= -1$

D Outcome Weights of $\hat{\tau}_{a,1}$

This section derives the outcome weights of $\hat{\tau}_{a,1}$ (also written $\hat{\tau}_t$ in ?) in the PIVE framework of ?. For a generic propensity-score estimate that does not impose the required balancing equality, the estimator is untreated-unnormalized.

Setup. Define the raw instrument IPW contrast for observation i as

$$r_i = \frac{Z_i}{\hat{p}_i} - \frac{1 - Z_i}{1 - \hat{p}_i} = \frac{Z_i - \hat{p}_i}{\hat{p}_i(1 - \hat{p}_i)}. \tag{105}$$

The two kappa components κ_{i1} and κ_{i0} decompose r_i as follows:

$$\kappa_{i1} = D_i \frac{Z_i - \hat{p}_i}{\hat{p}_i(1 - \hat{p}_i)} = D_i r_i, \qquad \kappa_{i0} = (1 - D_i)\frac{(1 - Z_i) - (1 - \hat{p}_i)}{\hat{p}_i(1 - \hat{p}_i)} = -(1 - D_i) r_i. \tag{106}$$

Equation (??) is the central identity: the treated part of r_i is κ_{i1} and the untreated part is $-\kappa_{i0}$.

PIVE representation. Define the diagonal transformation matrix $\mathbf{T}^{a,1} = \operatorname{diag}(r_1, \ldots, r_N)$. Using (??), the pseudo-treatment is $\mathbf{T}^{a,1}\mathbf{D} = \boldsymbol{\kappa}_1$, and the estimator can be written as

$$\hat{\tau}_{a,1} = \frac{N^{-1}\mathbf{1}_N' \mathbf{T}^{a,1}\mathbf{Y}}{N^{-1}\mathbf{1}_N' \mathbf{T}^{a,1}\mathbf{D}} = \frac{N^{-1}\sum_i r_i Y_i}{N^{-1}\sum_i \kappa_{i1}}, \tag{107}$$

which is a PIVE with $\widetilde{\mathbf{Z}} = \mathbf{1}_N$, $\widetilde{\mathbf{D}} = \boldsymbol{\kappa}_1$, $\widetilde{\mathbf{Z}}'\widetilde{\mathbf{D}} = \sum_i \kappa_{i1}$. Applying Equation (??), the outcome weights are

$$\boldsymbol{\omega}^{a,1\prime} = \left(\sum_i \kappa_{i1}\right)^{-1} \mathbf{1}_N' \mathbf{T}^{a,1}, \tag{108}$$

so that the weight of observation i is

$$\boxed{\omega_i^{a,1} = \frac{r_i}{\sum_j \kappa_{j1}} = \frac{1}{\sum_j \kappa_{j1}} \cdot \frac{Z_i - \hat{p}_i}{\hat{p}_i(1-\hat{p}_i)}.} \tag{109}$$

Normalization properties. *(i) Total weight sum.* Let $s_1 = \sum_i Z_i/\hat{p}_i$ and $s_0 = \sum_i(1 - Z_i)/(1-\hat{p}_i)$. Then $\sum_i r_i = s_1 - s_0$, so

$$\sum_{i=1}^{N} \omega_i^{a,1} = \frac{\sum_i r_i}{\sum_j \kappa_{j1}} = \frac{s_1 - s_0}{\sum_j \kappa_{j1}}, \tag{110}$$

Thus, $\sum_i \omega_i^{a,1} = 0$ if and only if $s_1 = s_0$. This equality is not imposed by a generic propensity-score estimator, so $\sum_i \omega_i^{a,1} \neq 0$ in general. The CBPS exception relevant for this thesis is stated in Remark **??** below. *(ii) Treated weight sum.* Using $D_i r_i = \kappa_{i1}$ from (**??**):

$$\sum_{i=1}^{N} \omega_i^{a,1} D_i = \frac{\sum_i D_i r_i}{\sum_j \kappa_{j1}} = \frac{\sum_i \kappa_{i1}}{\sum_j \kappa_{j1}} = 1. \qquad \checkmark \tag{111}$$

The treated weights sum to one exactly, because the denominator of $\hat{\tau}_{a,1}$ is $\sum_i \kappa_{i1}$ — precisely the treated-side kappa total.

(iii) Untreated weight sum. Using $(1-D_i)r_i = -\kappa_{i0}$ from (**??**):

$$\sum_{i=1}^{N} \omega_i^{a,1}(1-D_i) = \frac{\sum_i (1-D_i) r_i}{\sum_j \kappa_{j1}} = \frac{-\sum_i \kappa_{i0}}{\sum_j \kappa_{j1}}, \tag{112}$$

which equals -1 only if $\sum_i \kappa_{i0} = \sum_i \kappa_{i1}$. A generic propensity-score estimate does not enforce this finite-sample equality, and therefore $\sum_i \omega_i^{a,1}(1-D_i) \neq -1$ in general.

Classification. Collecting the three results:

$$\sum_{i=1}^{N} \omega_i^{a,1} \neq 0, \qquad \sum_{i=1}^{N} \omega_i^{a,1} D_i = 1, \qquad \sum_{i=1}^{N} \omega_i^{a,1}(1-D_i) = -\frac{\sum_i \kappa_{i0}}{\sum_i \kappa_{i1}} \neq -1. \tag{113}$$

Therefore, in the generic case, $\hat{\tau}_{a,1}$ is *untreated-unnormalized* in the sense of$\sum_i \kappa_{i0}$ for the two sides, forcing both to sum to their target values; $\hat{\tau}_{a,1}$ uses only the treated-side denominator, leaving the untreated side without its own rescaling. **?**, Table 4: the treated side is normalized by construction but the untreated side is not. The contrast with $\hat{\tau}_{a,10}$ is instructive: the normalized estimator uses separate denominators $\sum_i \kappa_{i1}$ and $\sum_i \kappa_{i0}$ for the two sides, forcing both to sum to their target values; $\hat{\tau}_{a,1}$ uses only the treated-side denominator, leaving the untreated side without its own rescaling.

Remark D.1 (Role of the separate Hájek normalization)**.** Equation (**??**) shows that the total outcome-weight mass of $\hat{\tau}_{a,1}$ remains proportional to the difference $s_1 - s_0$. By contrast, $\widehat{\tau}_u$ normalizes the two instrument groups separately, so Equation (**??**) reduces its total mass to $(\widehat{\Delta}_D^u)^{-1}(1-1)=0$. For this pair of Wald-IPW estimators, separate Hájek normalization is therefore the algebraic step that guarantees translation invariance without requiring an additional balancing restriction.

This comparison changes when the propensity-score procedure itself imposes that restriction. With an intercept included in the CBPS balancing equation (**??**),

$$s_1 - s_0 = \sum_{i=1}^{N} \frac{Z_i - \widehat{p}_i}{\widehat{p}_i(1-\widehat{p}_i)} = 0 \tag{114}$$

up to numerical tolerance. In this case, $\widehat{\tau}_{a,1}^{cb}$ also has zero total outcome-weight mass and coincides with $\widehat{\tau}_u^{cb}$, consistently with the finite-sample equivalence discussed in Chapter 3.

E Vietnam Application: Additional diagnostics

E.1 Draft eligibility ceilings

Table 13: Draft-eligibility ceilings by birth cohort (Vietnam application)

Birth cohort	Lottery year	Eligibility ceiling
1950	1970	$RSN_i \leq 195$
1951	1971	$RSN_i \leq 125$
1952	1972	$RSN_i \leq 95$
1953	1973	$RSN_i \leq 95$

E.2 Instrument Propensity Scores

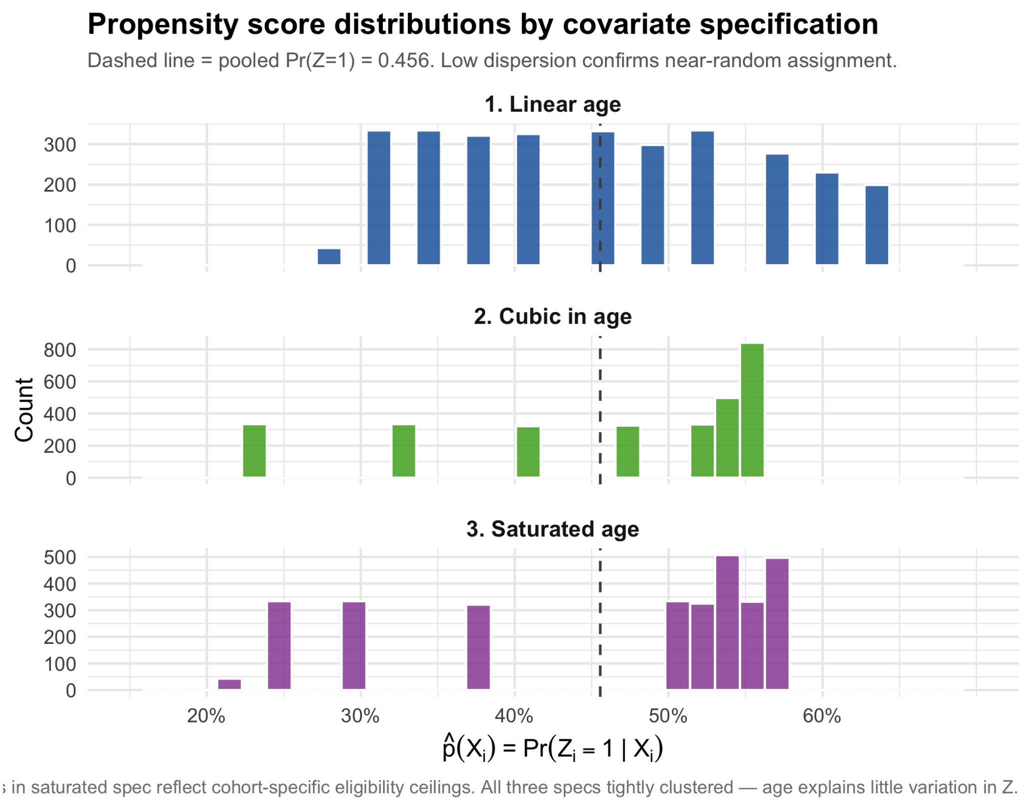

Figure 3: The first Empirical Application : Distribution of the fitted propensity score $\hat{p}(X_i) = \Pr(Z_i = 1 \mid X_i)$ under three covariate specifications.

E.3 Vietnam Design diagnostics

Table 14: Design diagnostics: Vietnam draft lottery

Quantity or specification	Estimate	Robust t^2
Sample size N	3,027	
$P(Z_i = 1)$	0.456	
$P(D_i = 1 \mid Z_i = 0)$	0.265	
$P(D_i = 1 \mid Z_i = 1)$	0.404	
Unadjusted uptake difference	0.139	
First stage, linear age	0.0949	30.1
First stage, cubic age	0.0882	26.3
First stage, saturated age	0.0899	27.2

Robust t^2 is the squared HC1-robust t-statistic for the excluded instrument.

E.4 Estimates kappa vs dml

Table 15: Point estimates for log hourly wages across age specifications

Estimator	Linear age	Cubic age	Saturated age
Panel A: 2SLS benchmark			
2SLS	0.233 (0.212)	0.227 (0.229)	0.254 (0.227)
Panel B: Normalised kappa estimators			
$\hat{\tau}_u^{cb}$	0.229 (0.213)	0.210 (0.233)	0.241 (0.229)
$\hat{\tau}_u^{ml}$	0.234 (0.211)	0.202 (0.235)	0.241 (0.229)
$\hat{\tau}_{a,10}$	0.227 (0.204)	0.204 (0.239)	0.241 (0.229)
Panel C: Unnormalised kappa estimators			
$\hat{\tau}_a$	0.015 (0.207)	0.314 (0.252)	0.241 (0.229)
$\hat{\tau}_t$	0.016 (0.219)	0.302 (0.240)	0.241 (0.229)
$\hat{\tau}_{a,0}$	0.014 (0.199)	0.317 (0.255)	0.241 (0.229)
Panel D: DML with honest-forest nuisances			
PLR-IV	n.a.	0.243 (0.228)	0.242 (0.228)
Wald-AIPW	n.a.	0.229 (0.231)	0.244 (0.233)

Notes: The outcome is log hourly wages measured in dollars. The 2SLS and kappa results correspond to the dollar specifications in the replication of Table 2 in **?**. The DML nuisance functions are estimated using tuned honest forests with five-fold cross-fitting.

E.5 Estimates flexible nuisance

Table 16: DML point estimates across nuisance learners

	PLR-IV		Wald-AIPW	
Nuisance learner	Cubic age	Saturated age	Cubic age	Saturated age
Honest forest	0.243 (0.228)	0.242 (0.228)	0.229 (0.231)	0.244 (0.233)
Regression	0.239 (0.231)	0.270 (0.230)	0.218 (0.233)	0.247 (0.234)
Random forest	0.270 (0.230)	0.283 (0.223)	0.246 (0.234)	0.261 (0.224)
XGBoost	0.265 (0.229)	0.281 (0.230)	0.238 (0.228)	0.253 (0.230)

Notes: The outcome is log hourly wages measured in dollars. All specifications use five-fold cross-fitting.

F Replication of the Słoczyński–Uysal–Wooldridge Vietnam results

Table 17: Replication of Słoczyński, Uysal, and Wooldridge (2025), Table 2

	Linear age		Cubic age		Saturated age	
Estimator	Cents	Dollars	Cents	Dollars	Cents	Dollars
Panel A: 2SLS benchmark						
2SLS	0.233	0.233	0.227	0.227	0.254	0.254
	(0.212)	(0.212)	(0.229)	(0.229)	(0.227)	(0.227)
Panel B: Normalised kappa estimators						
$\hat{\tau}_u^{cb}$	0.229	0.229	0.210†	0.210†	0.241	0.241
	(0.213)	(0.213)	(0.233)	(0.233)	(0.229)	(0.229)
$\hat{\tau}_u^{ml}$	0.234	0.234	0.202	0.202	0.241	0.241
	(0.211)	(0.211)	(0.235)	(0.235)	(0.229)	(0.229)
$\hat{\tau}_{a,10}$	0.227	0.227	0.204	0.204	0.241	0.241
	(0.204)	(0.204)	(0.239)	(0.239)	(0.229)	(0.229)
Panel C: Unnormalised kappa estimators						
$\hat{\tau}_a$	−0.429*	0.015	0.537*	0.314	0.241	0.241
	(0.258)	(0.207)	(0.322)	(0.252)	(0.229)	(0.229)
$\hat{\tau}_t$	−0.455	0.016	0.515*	0.302	0.241	0.241
	(0.279)	(0.219)	(0.301)	(0.240)	(0.229)	(0.229)
$\hat{\tau}_{a,0}$	−0.413*	0.014	0.540*	0.317	0.241	0.241
	(0.246)	(0.199)	(0.326)	(0.255)	(0.229)	(0.229)

Notes: Replication of Table 2 in Słoczyński, Uysal, and Wooldridge (2025). Log wages in cents equal log wages in dollars plus log(100). The specifications use linear age, a cubic polynomial in age, and age indicators, respectively. The instrument propensity score is estimated by logistic maximum likelihood, except for $\hat{\tau}_u^{cb}$, which uses CBPS balancing conditions. Standard errors are reported in parentheses and are HC1-robust for 2SLS and based on the M-estimation sandwich for the kappa estimators. The normalised estimates are unchanged across outcome units; the unnormalised estimates change under the linear and cubic specifications but coincide under saturated age indicators in this sample. † The published value is 0.208 with standard error 0.232; Section **??** explains the small numerical difference. $^*p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$.

F.1 Numerical difference in the cubic CBPS replication

The only difference from the published table is the cubic-age $\hat{\tau}_u^{cb}$ estimate. Both implementations use the same sample, covariates, estimator, and sandwich variance formula. The difference arises from the numerical solution of the CBPS balancing conditions in a poorly conditioned cubic polynomial. The Stata solution underlying the published value has a maximum absolute balancing residual of 5.45×10^{-4}, whereas the standardised R solver reduces this residual to 4.41×10^{-14}. Supplying the Stata propensity scores to the R estimator reproduces both the published estimate and its standard error. The analysis therefore retains the tightly balanced R result, 0.210 (0.233), while reporting the published value in the table note.

G Vietnam Outcome-Weight Diagnostics

Table 18: Complete outcome-weight diagnostics for the Vietnam application

Estimator	$\lvert\omega'Y - \hat{\tau}\rvert$	$\sum_i \omega_i$	Mass T/C	$\text{ESS}_{\text{mod}}/N$ T/C	Opposite-sign mass T/C	$\max_i \lvert\omega_i\rvert$
Panel A: Cubic age specification						
PLR-IV, tuned forest	6.88×10^{-15}	0.000000	0.994/0.994	0.963/0.921	43.5/46.6	0.012494
Wald-AIPW, tuned forest	8.69×10^{-15}	0.000000	1.000/1.000	0.952/0.910	43.4/46.7	0.017627
PLR-IV, tuned XGBoost	1.18×10^{-7}	0.000627	0.993/0.993	0.962/0.917	43.5/46.6	0.013060
Wald-AIPW, tuned XGBoost	3.99×10^{-9}	0.000027	1.001/1.001	0.951/0.907	43.3/46.7	0.019535
$\hat{\tau}_u^{cb}$	0	0.000000	1.000/1.000	0.958/0.908	43.5/46.8	0.025384
$\hat{\tau}_u^{ml}$	0	0.000000	1.000/1.000	0.954/0.899	43.6/46.8	0.028108
$\hat{\tau}_{a,10}$	0	0.000000	1.000/1.000	0.954/0.898	43.5/46.9	0.029078
$\hat{\tau}_a$	0	0.048243	1.041/0.993	0.954/0.898	43.5/46.9	0.028869
$\hat{\tau}_t$	0	0.046342	1.000/0.954	0.954/0.898	43.5/46.9	0.027731
$\hat{\tau}_{a,0}$	0	0.048594	1.049/1.000	0.954/0.898	43.5/46.9	0.029078
Panel B: Saturated age specification						
PLR-IV, tuned forest	5.33×10^{-15}	0.000000	0.999/0.999	0.961/0.917	43.5/46.6	0.013128
Wald-AIPW, tuned forest	5.97×10^{-15}	0.000000	1.005/1.005	0.951/0.908	43.4/46.7	0.016866
PLR-IV, tuned XGBoost	1.17×10^{-7}	0.000019	1.011/1.011	0.961/0.920	43.4/46.6	0.012257
Wald-AIPW, tuned XGBoost	3.46×10^{-8}	−0.000075	1.005/1.005	0.950/0.911	43.2/46.6	0.016570
$\hat{\tau}_u^{cb}$	0	0.000000	1.000/1.000	0.953/0.909	43.3/46.7	0.017811
$\hat{\tau}_u^{ml}$	0	0.000000	1.000/1.000	0.953/0.909	43.3/46.7	0.017811
$\hat{\tau}_{a,10}$	0	0.000000	1.000/1.000	0.953/0.909	43.3/46.7	0.017811
$\hat{\tau}_a$	0	0.000000	1.000/1.000	0.953/0.909	43.3/46.7	0.017811
$\hat{\tau}_t$	0	0.000000	1.000/1.000	0.953/0.909	43.3/46.7	0.017811
$\hat{\tau}_{a,0}$	0	0.000000	1.000/1.000	0.953/0.909	43.3/46.7	0.017811

Notes: T/C denotes treated and sign-oriented control values. The representation gap is the absolute difference between the weighted outcome sum and the reported estimate. The XGBoost rows use reconstructed weight vectors. Modified ESS is divided by the observed number of treated or control observations. Opposite-sign mass is reported in percent. The forest and XGBoost rows use the tuned specifications.

H Vietnam Covariate-Balance Diagnostics

Table 19: Learner-specific outcome-weight diagnostics: Vietnam draft lottery

Estimator	$\lvert\omega' Y - \hat{\tau}\rvert$	$\sum_i \omega_i$	Mass T/C	$\text{ESS}_{\text{mod}}/N$ T/C	Opposite-sign mass T/C	$\max_i \lvert\omega_i\rvert$
Panel A: Cubic age specification						
PLR-IV, OLS	3.71×10^{-9}	0.000000	1.000/1.000	0.966/0.924	43.6/46.7	0.014577
Wald-AIPW, OLS/logit	1.20×10^{-9}	0.000000	0.991/0.991	0.955/0.908	43.5/46.8	0.026926
PLR-IV, `ranger`	2.22×10^{-16}	0.000000	0.999/0.999	0.960/0.916	43.5/46.6	0.013538
Wald-AIPW, `ranger`	4.11×10^{-15}	0.000000	0.999/0.999	0.948/0.903	43.4/46.7	0.021647
Panel B: Saturated age specification						
PLR-IV, OLS	5.72×10^{-15}	0.000000	1.000/1.000	0.960/0.916	43.5/46.6	0.013426
Wald-AIPW, OLS/logit	3.37×10^{-14}	0.000000	1.000/1.000	0.948/0.903	43.4/46.7	0.021773
PLR-IV, `ranger`	2.11×10^{-15}	0.000000	1.003/1.003	0.964/0.924	43.3/46.5	0.011886
Wald-AIPW, `ranger`	3.77×10^{-15}	0.000000	1.005/1.005	0.951/0.914	43.1/46.6	0.016211

Notes: T/C denotes treated and sign-oriented control values. The representation gap is the absolute difference between the weighted outcome sum and the reported estimate. Modified ESS is divided by the observed number of treated or control observations. Opposite-sign mass is reported in percent.

pics/vietnam_love_cubic_thesis.png

Figure 4: Covariate balance under cubic age adjustment

Notes: Dark circles report the unadjusted absolute standardised mean differences. Yellow triangles report the differences after applying the estimator-specific outcome weights. The dashed vertical line marks 0.1. The XGBoost panels use the reconstructed tuned weight vectors. Balance is calculated with `cobalt` (**?**).

pics/vietnam_love_saturated_thesis.png

Figure 5: Covariate balance under saturated age adjustment

Notes: Dark circles report the unadjusted absolute differences in the binary age indicators. Yellow triangles report the differences after applying the estimator-specific outcome weights. The dashed vertical line marks 0.1. The XGBoost panels use the reconstructed tuned weight vectors. Balance is calculated with `cobalt` (**?**).

I Card Application: Additional Results and Diagnostics

I.1 Variables and Covariate Specifications

Table 20: Instrument, treatments, and outcome in the Card application

Role	Symbol	Definition
Instrument	Z_i	Grew up near a four-year college in 1966
Treatment 1	D_{1i}	Schooling beyond high school, $\mathbf{1}\{\texttt{educ}_i > 12\}$
Treatment 2	D_{2i}	At least sixteen years of schooling, $\mathbf{1}\{\texttt{educ}_i \geq 16\}$
Outcome	Y_i	Log hourly wage

Table 21: Covariate specifications in the Card application

Covariate	Card specification	Kitagawa specification
Experience and experience squared	Yes	No
Black indicator	Yes	Yes
Southern residence in 1976	Yes	Yes
Metropolitan residence in 1976	Yes	Yes
Metropolitan residence in 1966	Yes	Yes
Southern residence in 1966	No	Yes
Regional indicators, `reg661`, ..., `reg668`	Yes	No

I.2 Instrument-Propensity Support

Table 22: Fitted instrument-propensity support in the Card application

Specification	Fitted range	SD	Max., $Z = 0$	$Z = 0$ $\hat{p} > .90$	Min., $Z = 1$	$Z = 1$ $\hat{p} < .10$
Card	$[0.1724, 0.9517]$	0.2356	0.9357	40	0.1724	0
Kitagawa	$[0.2532, 0.9067]$	0.2280	0.9067	9	0.2532	0

Notes: The analysis sample contains 957 observations with $Z = 0$ and 2,053 observations with $Z = 1$. The last four columns describe the instrument-group tails that enter inverse instrument-propensity factors.

pics/card_propensity_support.png

Figure 6: Distribution of the fitted instrument propensity score by instrument status

Notes: The panels show the Card and Kitagawa covariate specifications. Each panel separates observaftions with $Z = 1$ and $Z = 0$. The dashed line marks the pooled instrument share.

I.3 Additional Point Estimates

Table 23: Full set of kappa estimates in the Card application

	Schooling beyond high school		At least 16 years	
Estimator	Card	Kitagawa	Card	Kitagawa
$\hat{\tau}_u^{cb}$	0.376 (0.223)	0.331 (0.236)	0.853 (0.549)	0.588 (0.433)
$\hat{\tau}_u^{ml}$	0.331 (0.202)	0.356 (0.244)	0.619 (0.387)	0.628 (0.448)
$\hat{\tau}_{a,10}^{ml}$	0.346 (0.200)	0.293 (0.252)	0.586 (0.356)	0.836 (0.821)
$\hat{\tau}_a^{ml}$	0.170 (0.370)	0.842 (0.362)	0.315 (0.696)	1.617 (0.891)
$\hat{\tau}_t^{ml}$	0.171 (0.367)	0.769 (0.308)	0.319 (0.687)	1.367 (0.648)
$\hat{\tau}_{a,0}^{ml}$	0.154 (0.354)	1.066 (0.574)	0.266 (0.639)	2.712 (2.577)

Notes: Each cell reports the point estimate followed by its standard error in parentheses. The first three rows are normalised and the last three rows are unnormalised.

Table 24: DML learner sensitivity in the Card application

	Schooling beyond high school		At least 16 years	
Estimator and learner	Card	Kitagawa	Card	Kitagawa
PLR-IV, linear	0.678	0.568	1.419	0.943
PLR-IV, ranger	0.685	0.623	1.112	1.045
PLR-IV, tuned XGBoost	0.577	0.612	0.956	1.010
Wald-AIPW, linear/logit	0.399	0.311	0.884	0.534
Wald-AIPW, ranger	0.262	0.294	0.493	0.514
Wald-AIPW, tuned XGBoost	0.382	0.238	0.730	0.477

Notes: The table reports point estimates. The honest-forest results are reported in Table **??**. XGBoost uses fold-specific nested tuning based only on nuisance-prediction loss. The untuned and tuned XGBoost comparison is documented separately in the computational appendix.

I.4 Complete-Pipeline Translation-Invariance Results

pics/card_translation_invariance.png

Figure 7: Complete-pipeline translation-invariance rerun in the Card application

Notes: The horizontal axis shows the difference between the estimate after the outcome translation and the original estimate. A symmetric pseudo-log scale displays small numerical differences and larger changes in the same figure. The dashed lines mark the tolerance of $\pm 10^{-4}$. The normalised and unnormalised kappa rows show $\hat{\tau}_u^{ml}$ and $\hat{\tau}_t^{ml}$.

Table 25: Translation-invariance rerun differences in the Card application

	Schooling beyond high school		At least 16 years	
Estimator	Card	Kitagawa	Card	Kitagawa
$\hat{\tau}_u^{cb}$	0.000000	0.000000	0.000000	0.000000
$\hat{\tau}_u^{ml}$	0.000000	0.000000	0.000000	0.000000
$\hat{\tau}_{a,10}^{ml}$	0.000000	0.000000	0.000000	0.000000
$\hat{\tau}_a^{ml}$	-0.488839	1.405803	-0.909273	2.700096
$\hat{\tau}_t^{ml}$	-0.492106	1.283819	-0.920642	2.283386
$\hat{\tau}_{a,0}^{ml}$	-0.444597	1.780059	-0.767256	4.529008
PLR-IV, tuned forest	-0.000556	0.000014	-0.000931	0.000025
Wald-AIPW, tuned forest	-0.001706	0.000050	-0.003239	0.000091
PLR-IV, linear	0.000000	0.000000	0.000000	0.000000
Wald-AIPW, linear/logit	0.000000	0.000000	0.000000	0.000000
PLR-IV, ranger	-0.000608	0.000010	-0.000987	0.000017
Wald-AIPW, ranger	-0.000351	0.000003	-0.000659	0.000005
PLR-IV, tuned XGBoost	-0.013188	0.000319	-0.021826	0.000527
Wald-AIPW, tuned XGBoost	-0.000229	0.004216	-0.000438	0.008437

Notes: Each entry is $\hat{\tau}(Y + \log(100)) - \hat{\tau}(Y)$ after repeating the complete estimation procedure. Absolute differences no larger than 10^{-4} are classified as numerical equality. Values displayed as zero are zero up to the reported precision.

I.5 Outcome-Weight Normalisation and Concentration

Table 26: Selected outcome-weight diagnostics in the Card application

Cell	Estimator	Treated/control mass	ESS_{mod}	$\max_i \lvert\omega_i\rvert$
Card, D_1	PLR-IV, tuned forest	1.00/1.00	1819	0.033
	Wald-AIPW, tuned forest	0.97/0.97	1345	0.146
	$\hat{\tau}_u^{ml}$	1.00/1.00	1569	0.052
	$\hat{\tau}_t^{ml}$	1.00/1.11	1561	0.055
Card, D_2	PLR-IV, tuned forest	0.96/0.96	1819	0.055
	Wald-AIPW, tuned forest	0.94/0.94	1345	0.277
	$\hat{\tau}_u^{ml}$	1.00/1.00	1569	0.097
	$\hat{\tau}_t^{ml}$	1.00/1.20	1561	0.103
Kitagawa, D_1	PLR-IV, tuned forest	1.00/1.00	1954	0.033
	Wald-AIPW, tuned forest	0.99/0.99	1631	0.065
	$\hat{\tau}_u^{ml}$	1.00/1.00	1764	0.038
	$\hat{\tau}_t^{ml}$	1.00/0.72	1786	0.038
Kitagawa, D_2	PLR-IV, tuned forest	1.01/1.01	1954	0.057
	Wald-AIPW, tuned forest	1.01/1.01	1631	0.119
	$\hat{\tau}_u^{ml}$	1.00/1.00	1764	0.078
	$\hat{\tau}_t^{ml}$	1.00/0.50	1786	0.067

Notes: Control mass is the negative of the sum of control-group weights, so a fully normalised treated-minus-control comparison has mass 1/1. Modified effective sample size is used because the outcome weights are signed. The table reports selected estimators needed to interpret the translation result.

I.6 Covariate Balance

pics/card_balance_card_educ16.png

Figure 8: Outcome-weight balance under the Card specification for the sixteen-year margin

Notes: Dark circles show unadjusted absolute standardised differences between the two treatment groups. Yellow triangles show the corresponding differences after applying the estimator-specific outcome weights. The dashed line marks 0.1. The figure compares untuned and tuned honest forests with the three normalised kappa estimators.

pics/card_balance_learners_card_educ16.png

Figure 9: Learner sensitivity of outcome-weight balance under the Card specification

Notes: The treatment is the sixteen-year schooling margin. Dark circles show unadjusted absolute standardised differences and yellow triangles show differences after weighting. XGBoost uses the tuned specification. These are balance diagnostics for the weighted treatment contrast. They are not tests of the conditional independence or exclusion restrictions.

J Subsampling Procedure for the Angrist and Evans Application

The diagnostic subsamples used in the Angrist–Evans application are constructed by proportional stratified subsampling without replacement. Let Z_i denote the same-sex instrument and D_i the more-than-two-children treatment. For the labor-force participation sample, the strata are defined by Z_i, D_i, the binary labor-force outcome, positive-income status, race, and mother's age group. For the log-income sample, the strata are defined by Z_i, D_i, income decile, race, and mother's age group. For each parent sample, let N_0 denote the parent-sample size and let n_s denote the size of stratum $s = 1, \ldots, S$. The target subsample size is set to $N = 3{,}000$. The proportional target allocation for each stratum is

$$n_s^* = N \frac{n_s}{N_0}.$$

The preliminary allocation is obtained by taking the integer part, $\lfloor n_s^* \rfloor$. The remaining observations,

$$N - \sum_{s=1}^{S} \lfloor n_s^* \rfloor,$$

are then assigned one at a time to the strata with the largest fractional remainders, until the allocated total equals N exactly. Within each stratum, the allocated number of observations is drawn randomly without replacement. The procedure is implemented separately for the labor-force participation and log-income parent samples, using a fixed seed. This yields two diagnostic subsamples of 3,000 observations each.

K Covariate Balance and Child-Sex Indicators

pics/ae_raw_covariate_dist.png

Figure 10: Covariate Balance by instrument status of having sam sex or not